From Image Classification to Dense Prediction

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University of Colorado Boulder
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https://dannagurari.colorado.edu/course/neural-networks-and-deep-learning-spring-2026/

Review

- Last lecture:
  - The quest for more data and faster machines
  - Paradigm shift: learned representations thrive with large labelled datasets
  - Lucky break: gradient descent meets gaming hardware
  - Tipping point: open-source platforms catalyze widespread engagement
  - Programming tutorial
- Assignments (Canvas)
  - Problem set 1 grades were released on Friday
  - Lab assignment 2 due Friday
    - For clarifications about common questions, review the Canvas assignment page
- Questions?
  - Beyond questions for me, please post them to Piazza and/or attend office hours

Today’s Topics

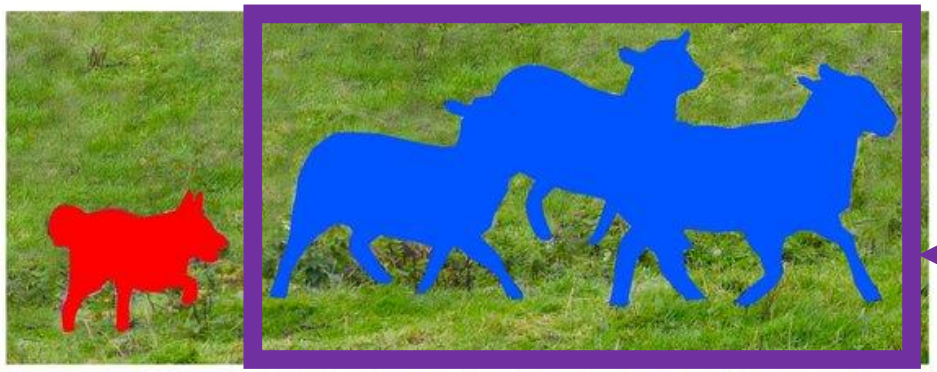
- Motivation: when vision asks not just “what”, but also “where”
- Redesigning neural networks to predict densely, rather than just classify
- Freeing dense prediction networks from fixed input sizes
- Pivotal insights: pretrained features + design tricks to boost spatial precision
- Programming tutorial

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Dense Prediction Tasks

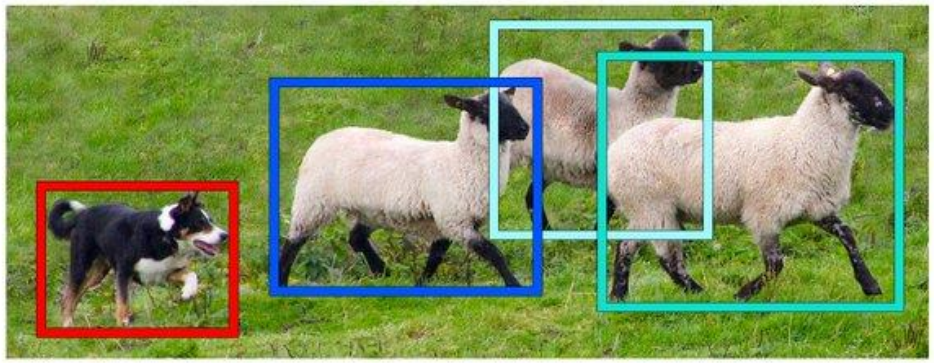
Locate all pixels belonging to pre-specified categories



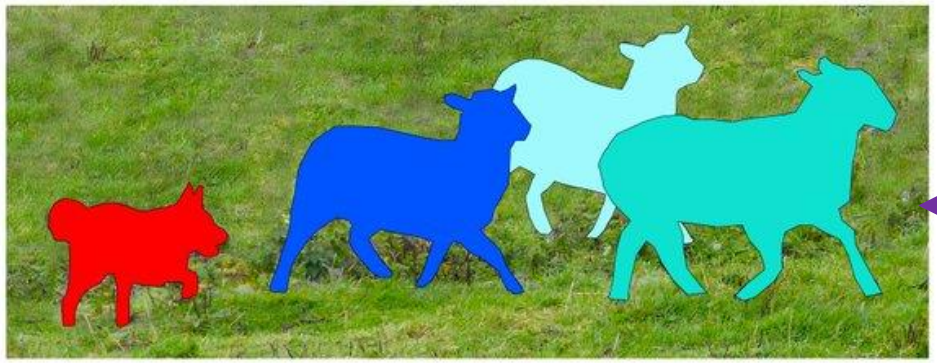
Semantic Segmentation

Note: instances of same category NOT separated

Locate all instances belonging to pre-specified categories with rectangles (aka, bounding boxes)



Object Detection

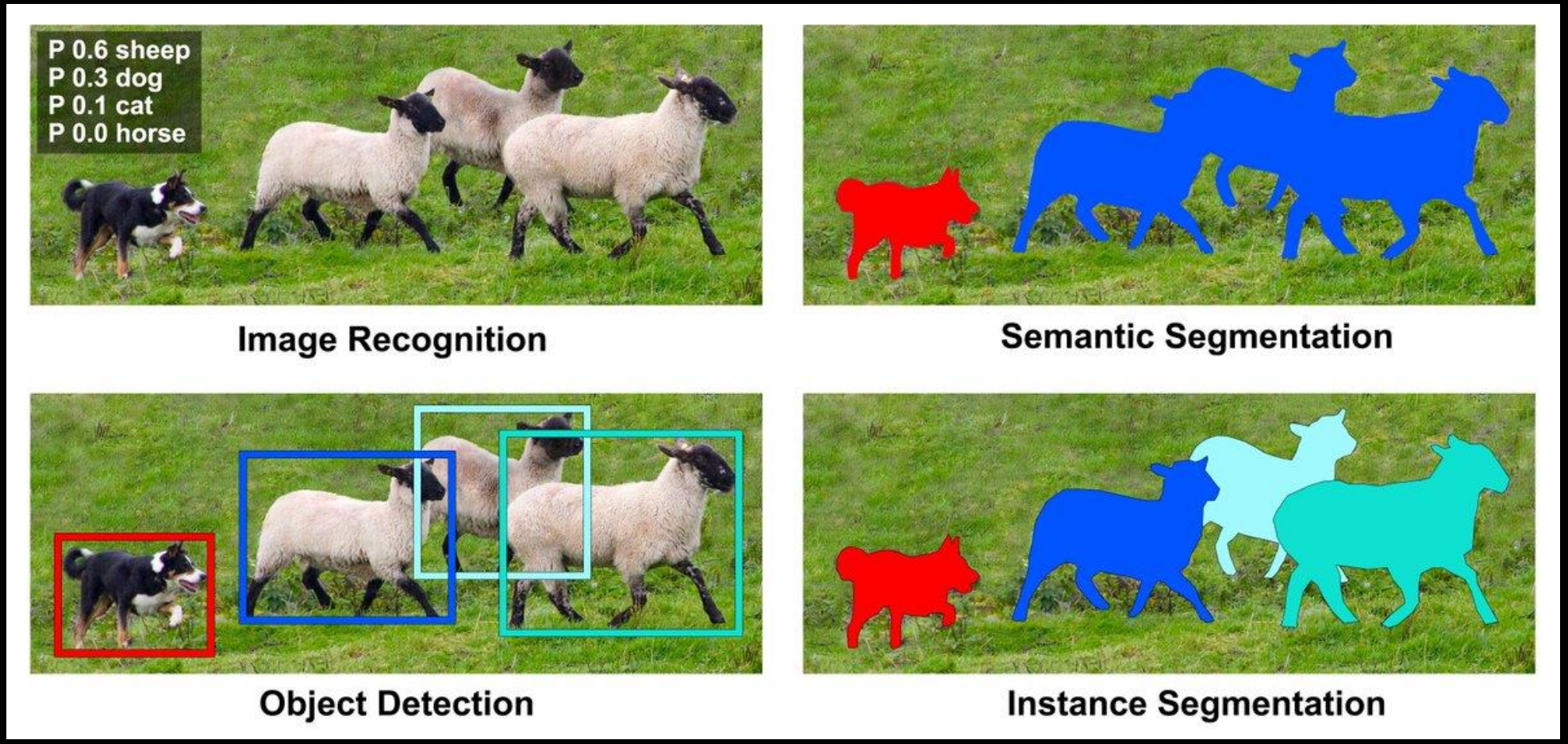


Instance Segmentation

Unifies semantic segmentation and object detection

https://ai-pool.com/d/could-you-explain-me-how-instance-segmentation-works

Object Recognition vs Dense Prediction



How does dense prediction differ from object recognition?

https://ai-pool.com/d/could-you-explain-me-how-instance-segmentation-works

Object Recognition vs Dense Prediction

How do failure modes of dense prediction differ from those of object recognition? e.g.,



MUST learn objects' appearances, rather than only image context;
e.g., common bias: if see savannah-like landscape, predict "giraffe"

Many Motivating Applications; e.g.,

Rotoscoping



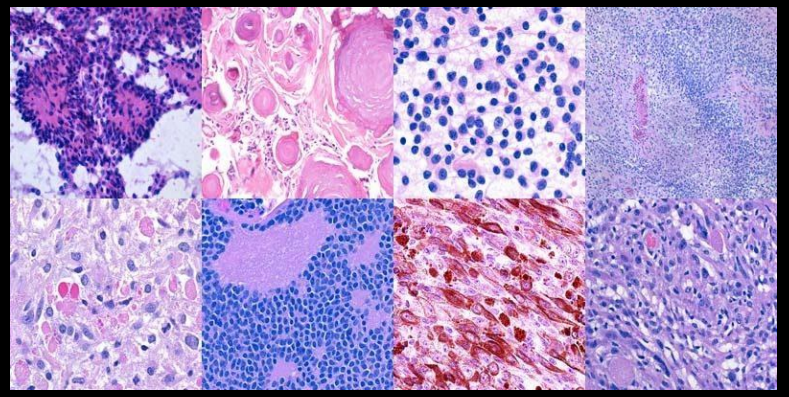
https://www.starnow.co.uk/ahmedmohammed1/photos/4650871/before-and-after-rotoscopinggreen-screening

Remodeling



Bell et al; SIGGRAPH; 2013

Disease Diagnosis



https://pathology.jhu.edu/brain-tumor/grading-classification

Automatic License Plate Recognition



Self-driving Cars



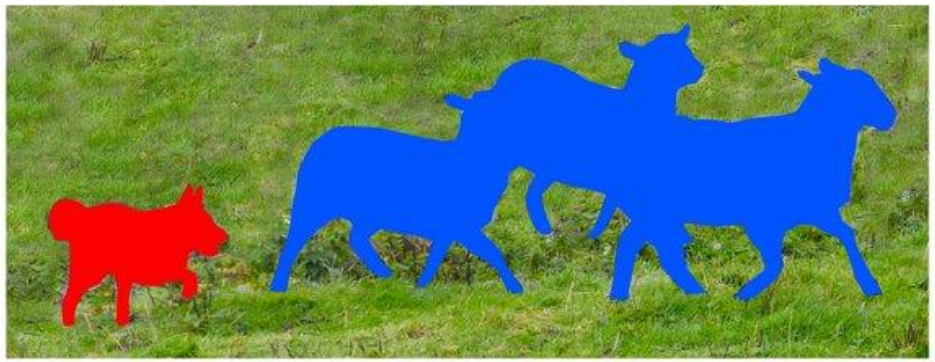
https://www.inc.com/kevin-j-ryan/self-driving-cars-powered-by-people-playing-games-mighty-ai.html

Common Pre-NN Approach: Inefficient Sliding Window (Repeated for Each Size/Scale)



https://yourboulder.com/boulder-neighborhood-downtown/

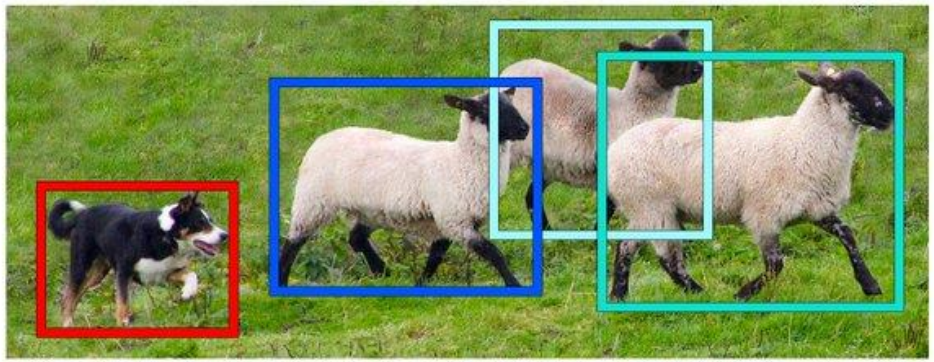
Pioneering Neural Networks for Dense Prediction (All Achieved State-of-the-Art Performance)



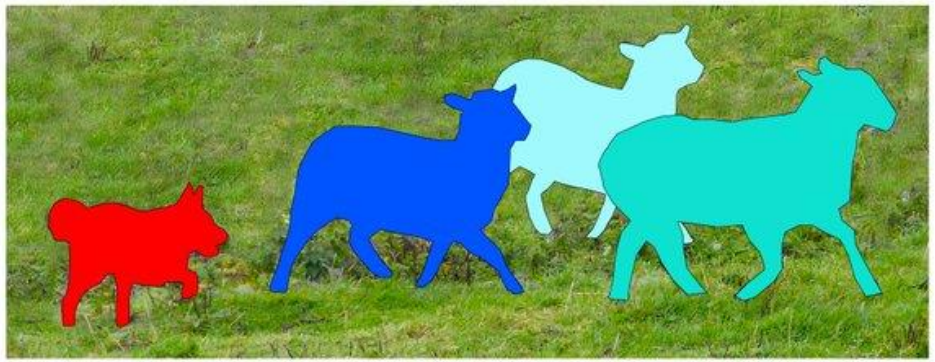
Semantic Segmentation

FCN in 2015

Faster R-CNN in 2015



Object Detection



Instance Segmentation

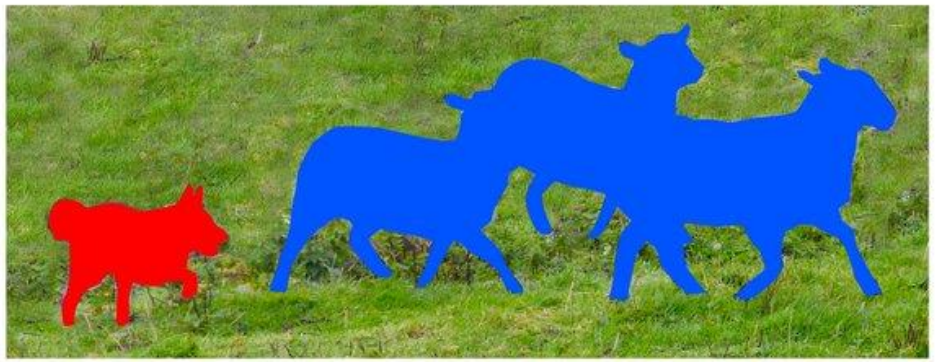
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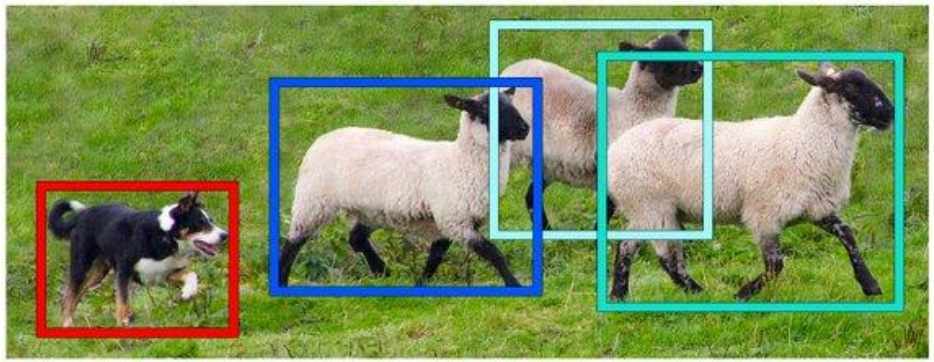
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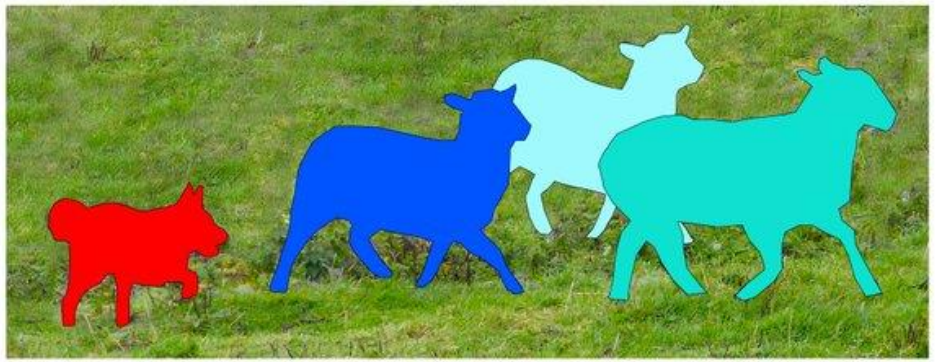
Ways to Change Networks' Outputs



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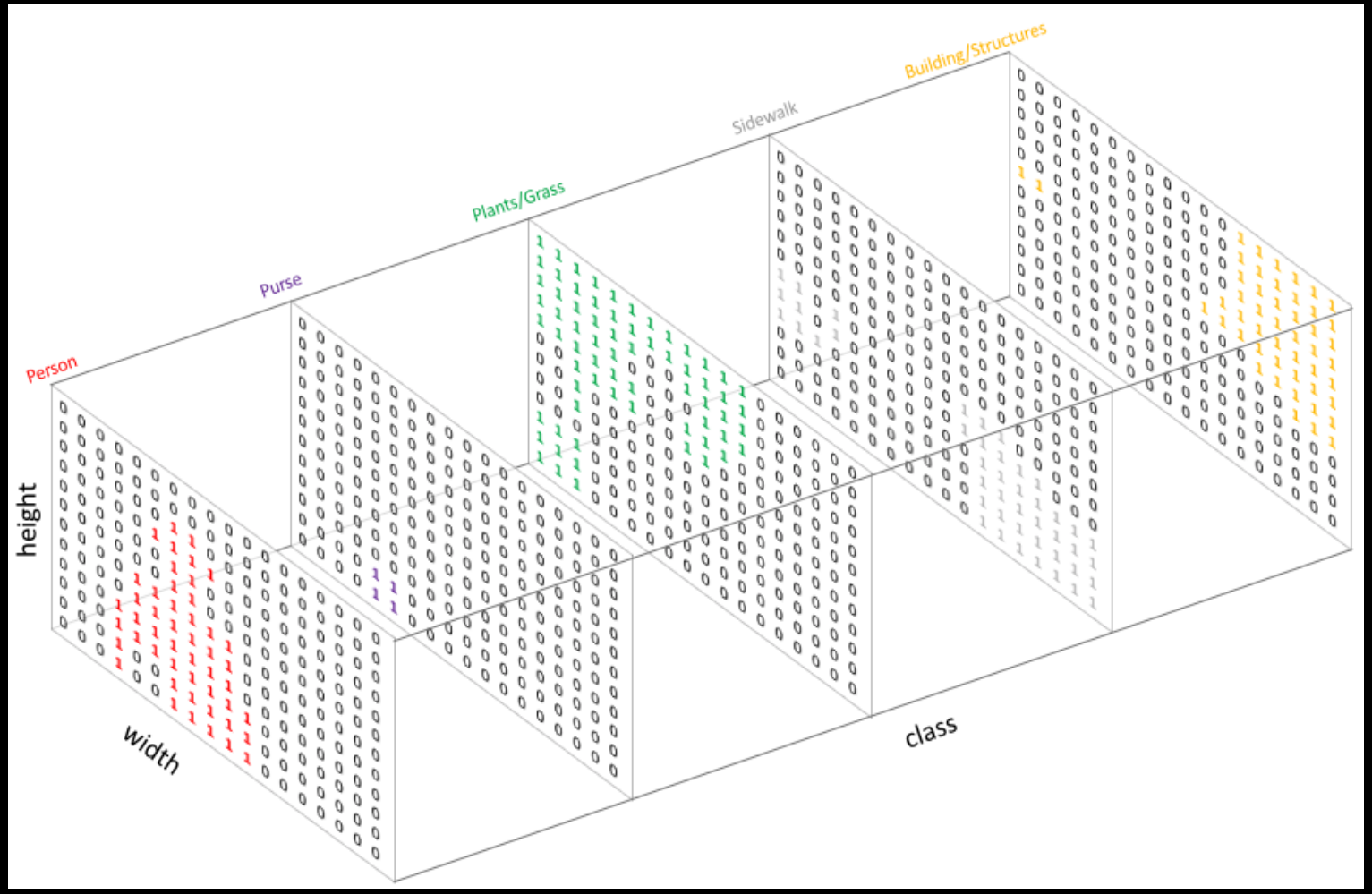
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Key Idea (1 of 2): Output Layer with Per-Pixel Classification

- Architecture: softmax generates a probability distribution over all categories for EVERY pixel

- In this example, what is the category for the:
  - upper left pixel?
  - bottom right pixel?

- Training: sums cross-entropy loss for EVERY pixel (acts like mini-batch for learning)

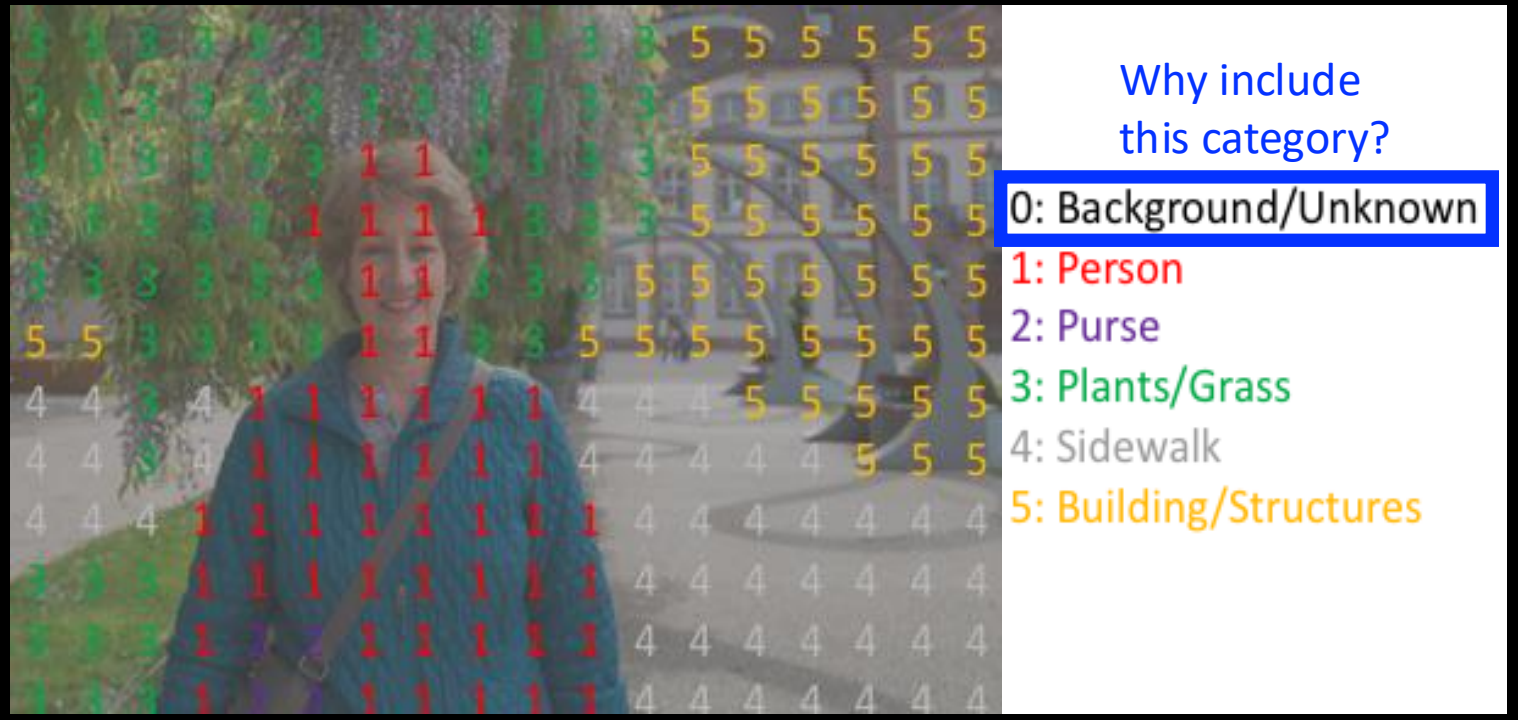


https://www.jeremyjordan.me/semantic-segmentation/

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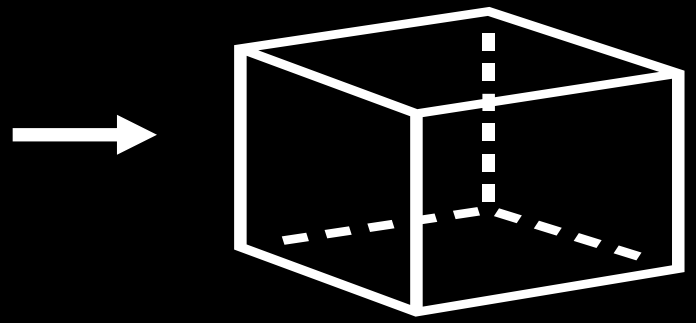
- How many categories can be predicted in this example?



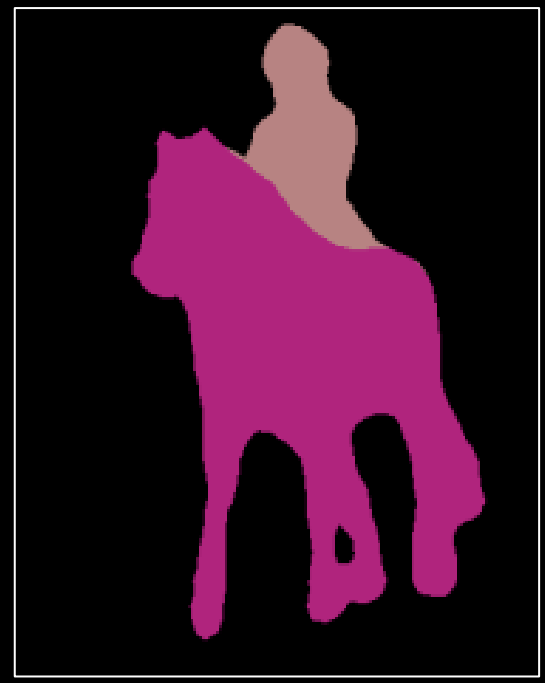
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Key Idea (2 of 2): Image Translation with Encoder-Decoder Architecture



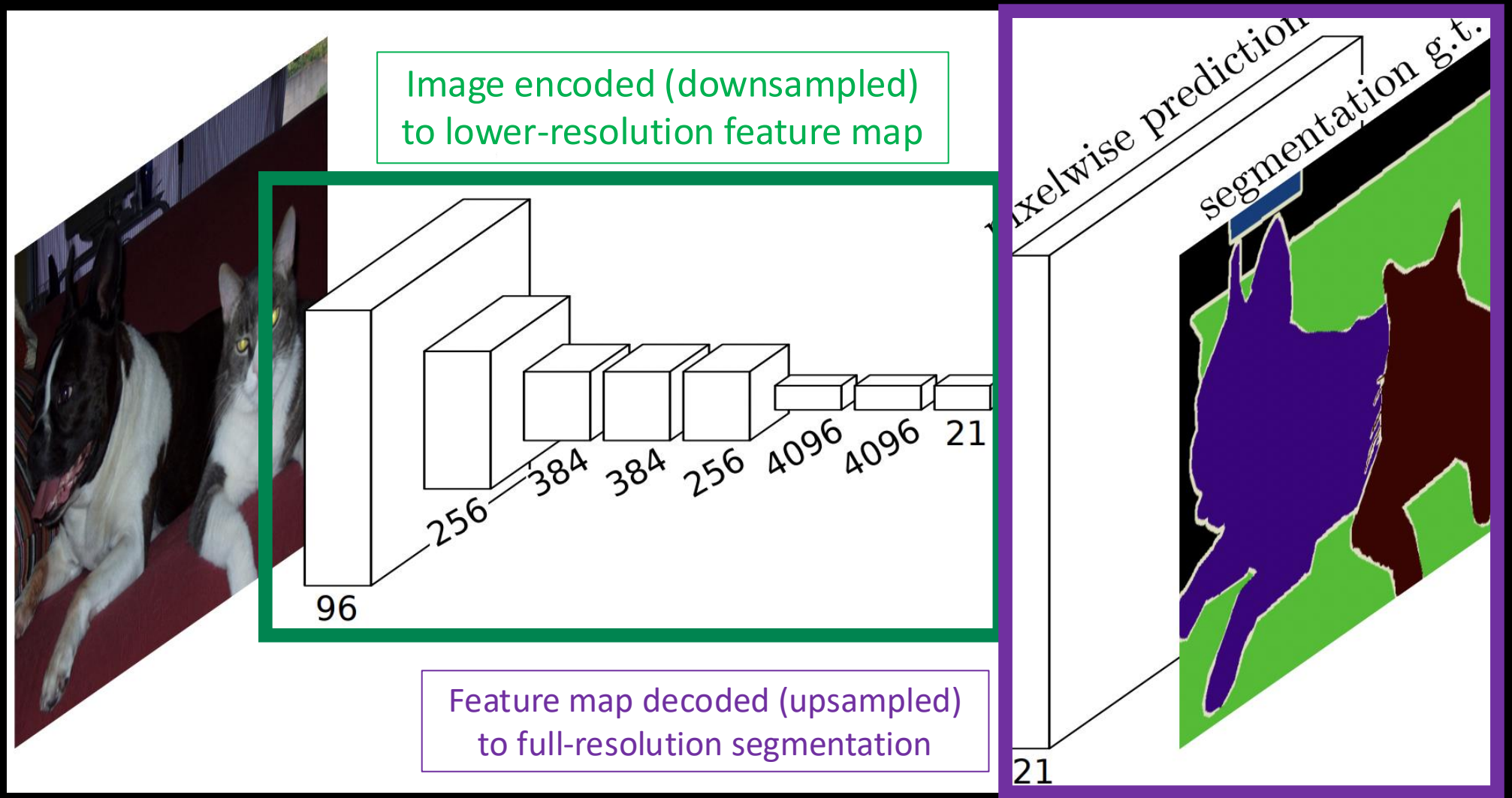


How is image resolution maintained?



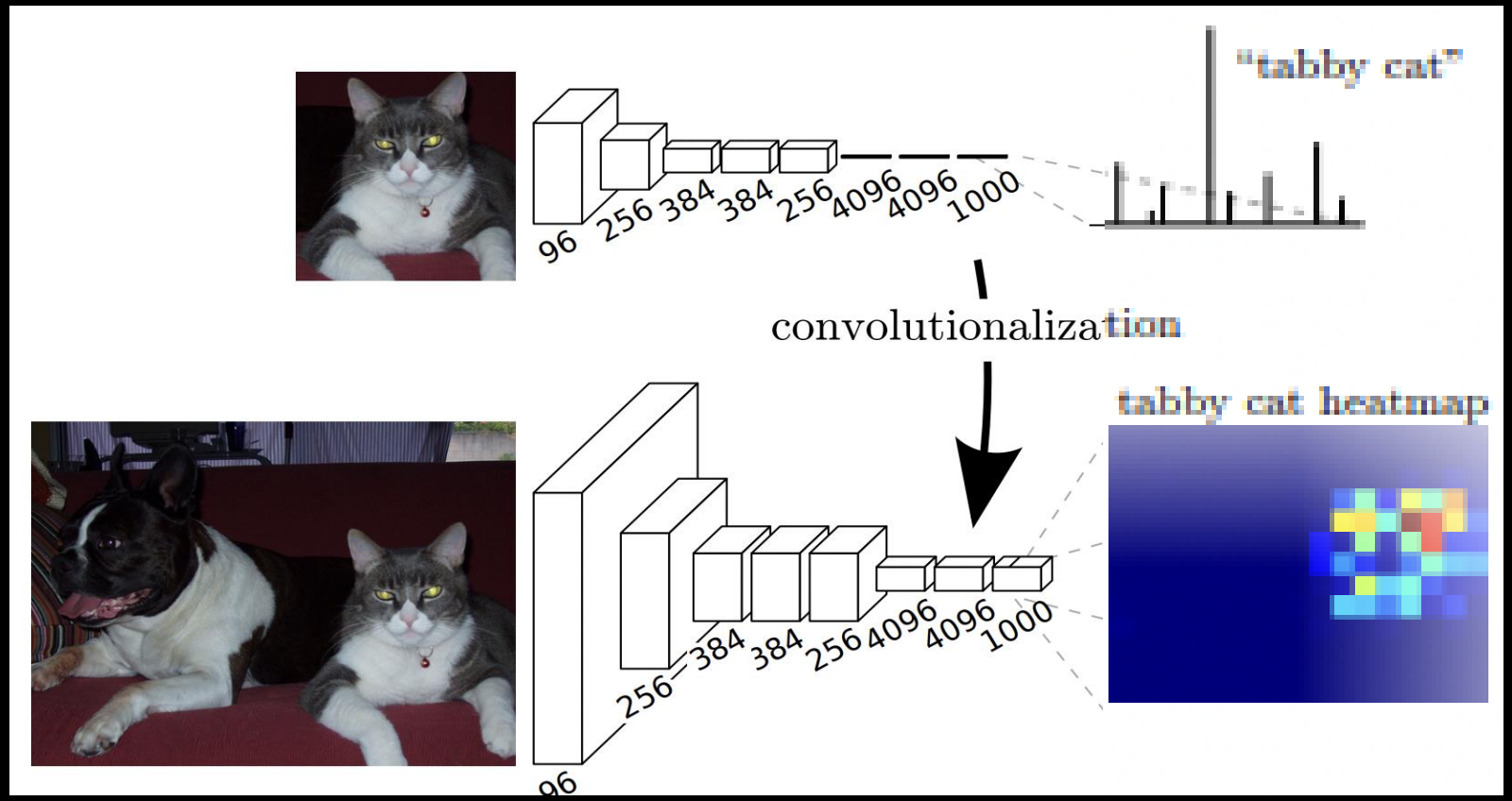
Long, Shelhamer, and Darrell. Fully Convolutional Networks for Semantic Segmentation. CVPR 2015.

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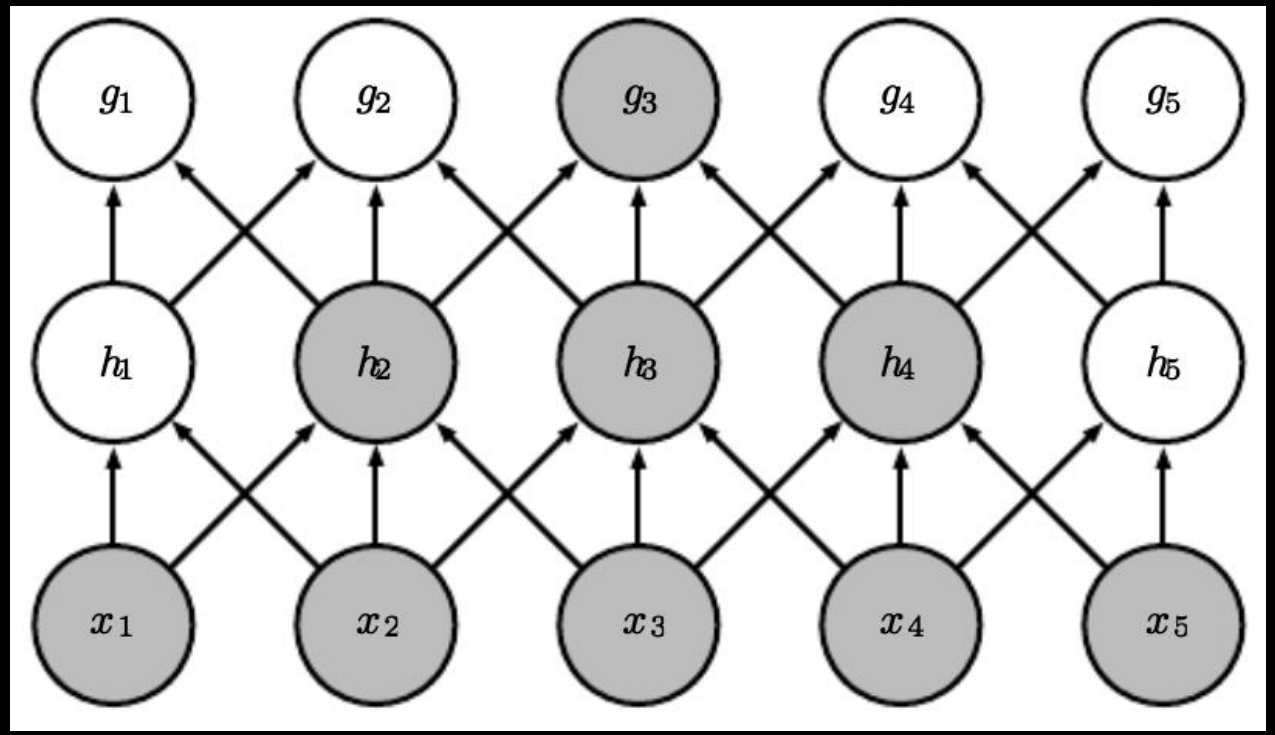
Fully connected layers requiring fixed number of weights replaced with convolutional layers to output a slice/heatmap per class for coarse image regions

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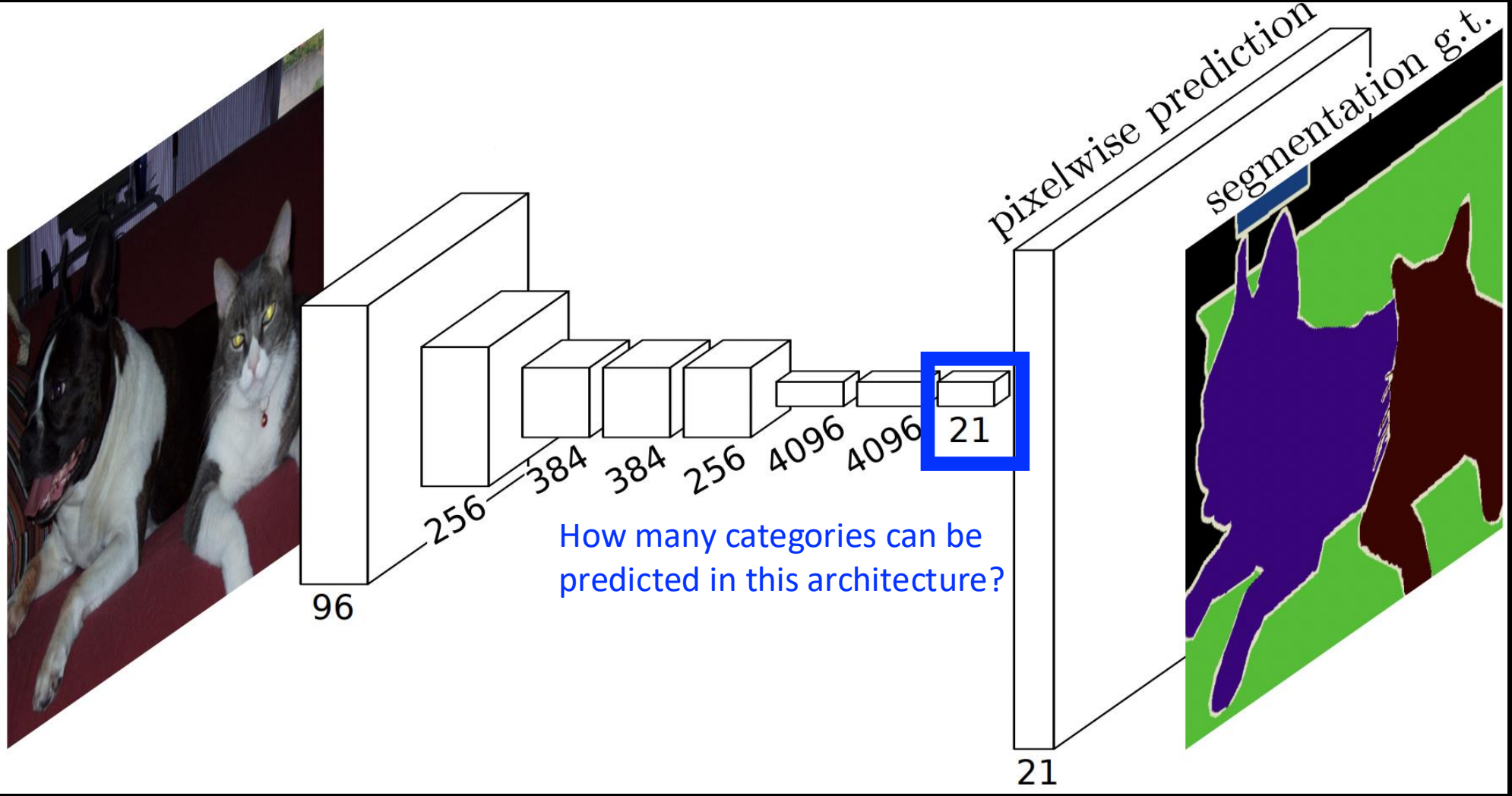
Why is this coarse region classification?

Recall stacked convolutions learn patterns over progressively **larger regions of the input (e.g., pixel) space**



https://www.deeplearningbook.org/contents/convnets.html

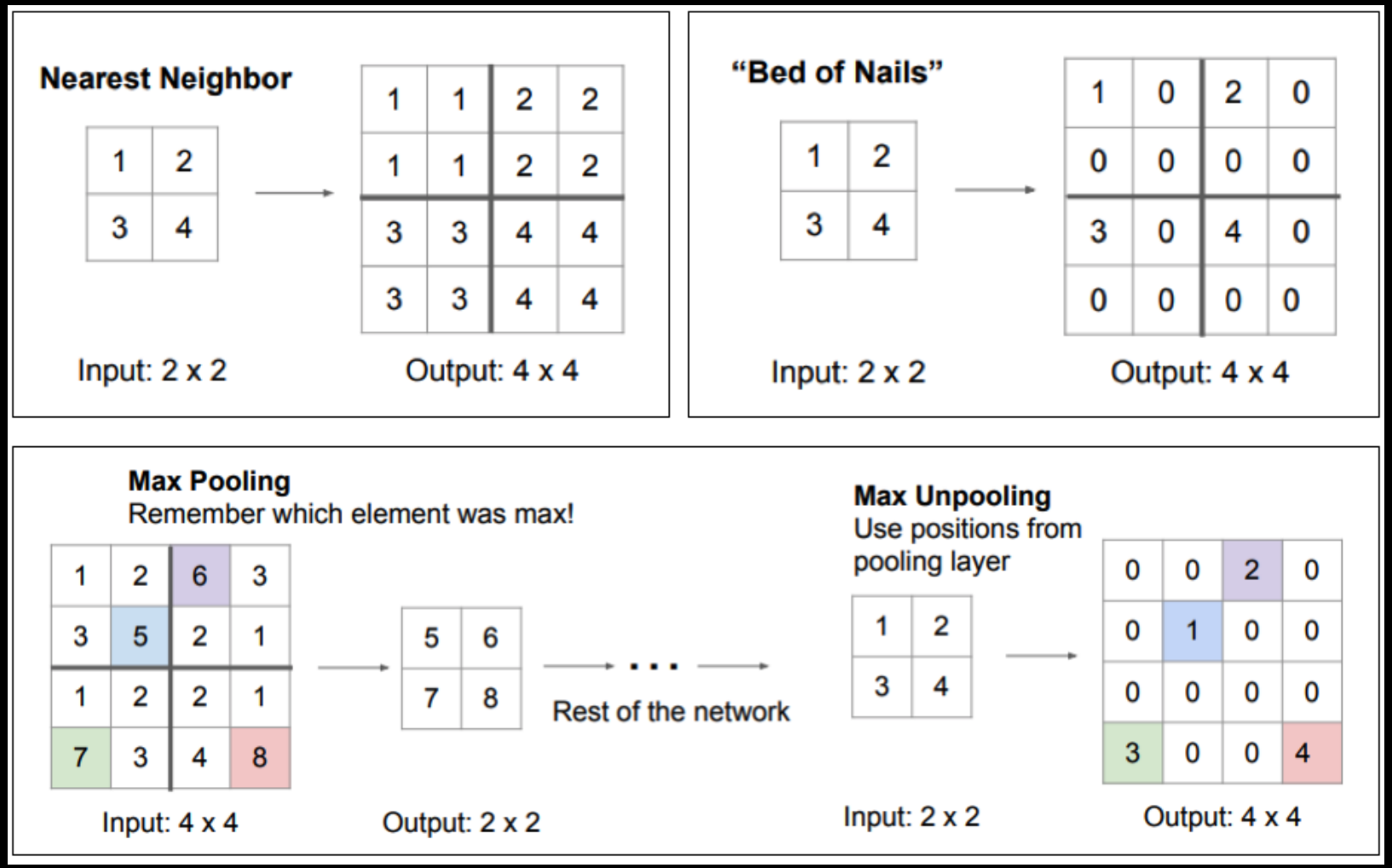
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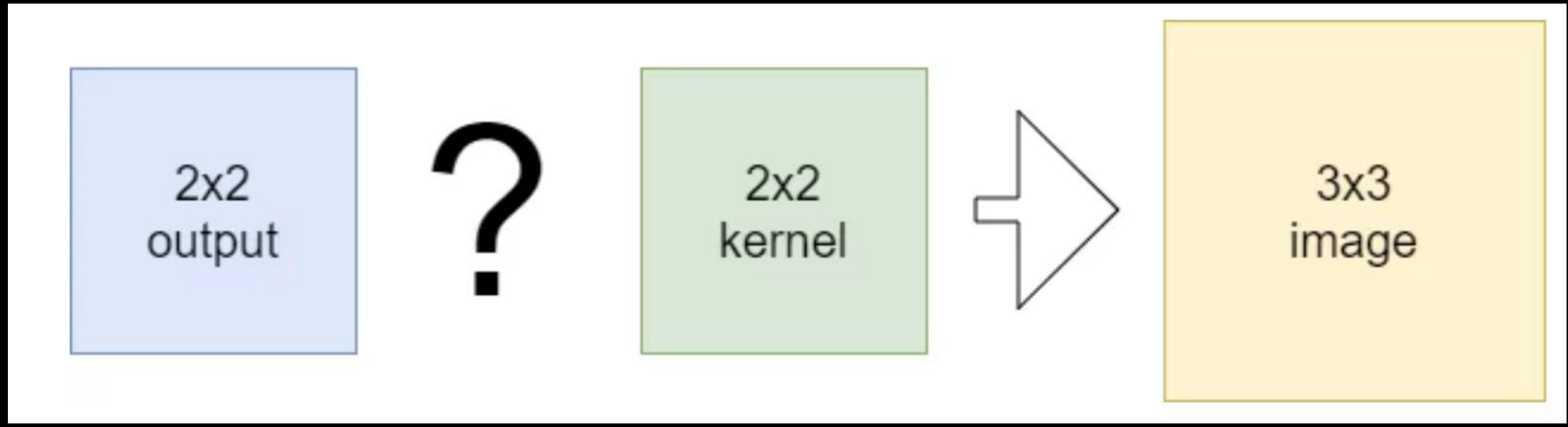
Many options for upsampling; e.g.,



http://cs231n.stanford.edu/slides/2017/cs231n_2017_lecture11.pdf

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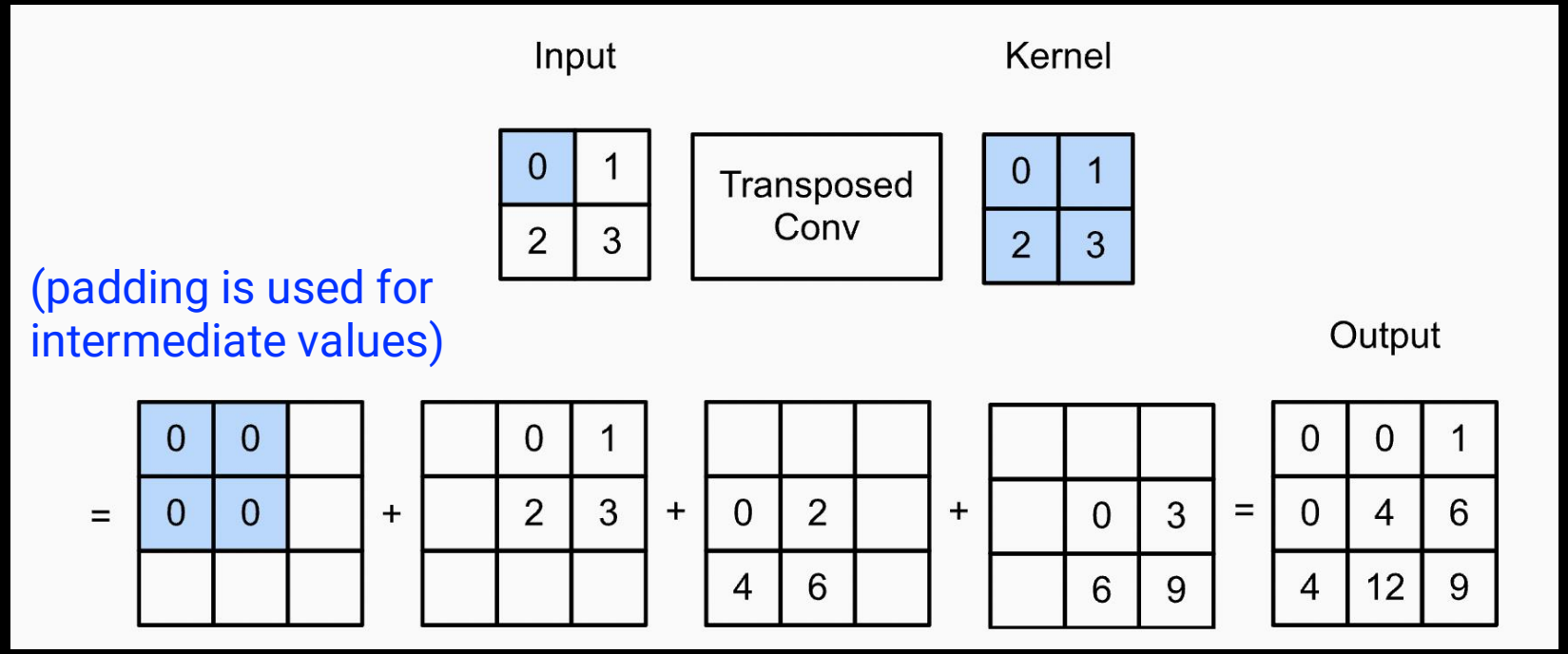
Many options for upsampling; e.g., **transposed convolutional layer** that learn filters that upsample the input with “fractional” sized steps
- also called “fractional convolutional layer”, “backward convolution”, and, incorrectly, ”deconvolution layer”



https://www.machinecurve.com/index.php/2019/09/29/understanding-transposed-convolutions/#the-goal-reconstructing-the-original-input

Key Idea (2 of 2): Image Translation with Encoder-Decoder Architecture

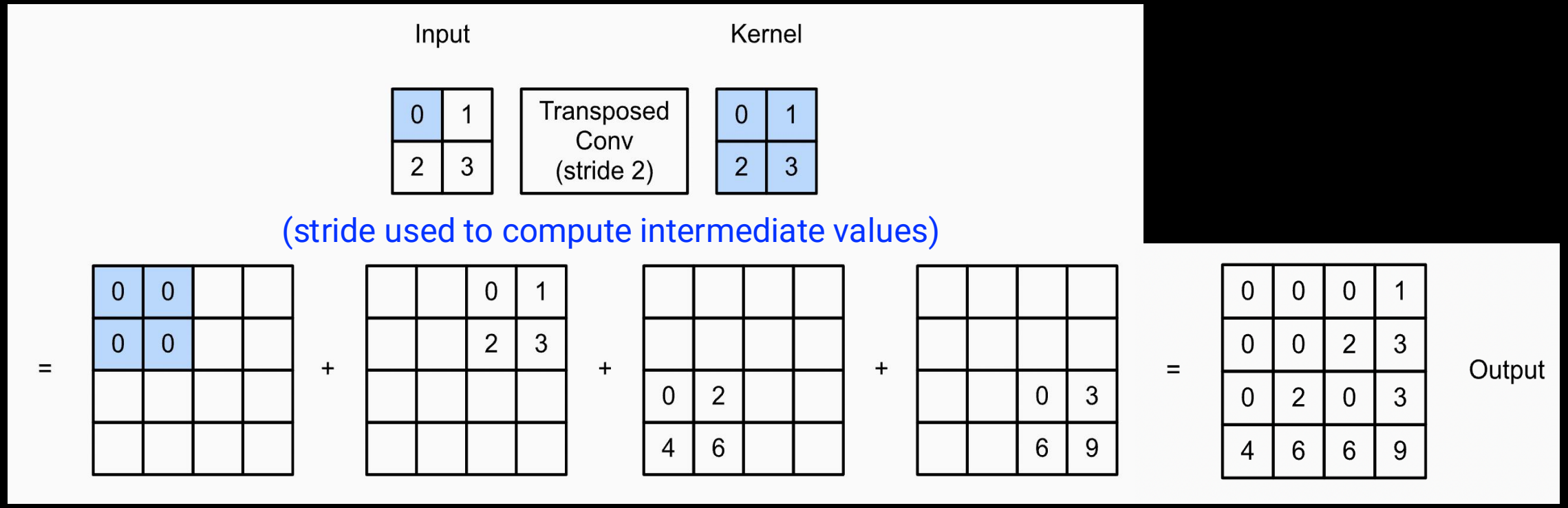
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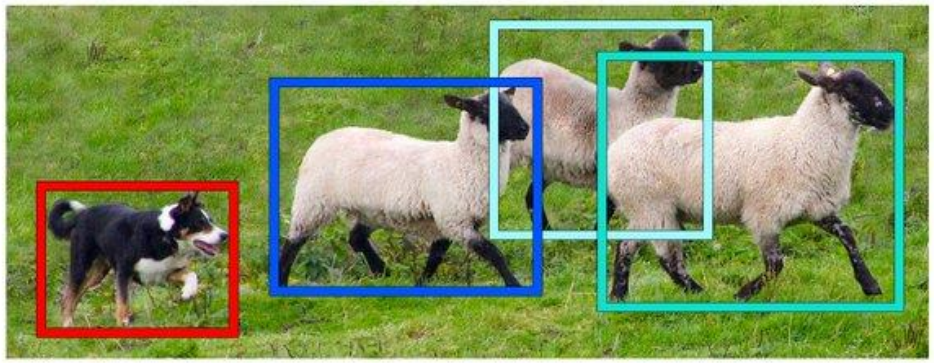
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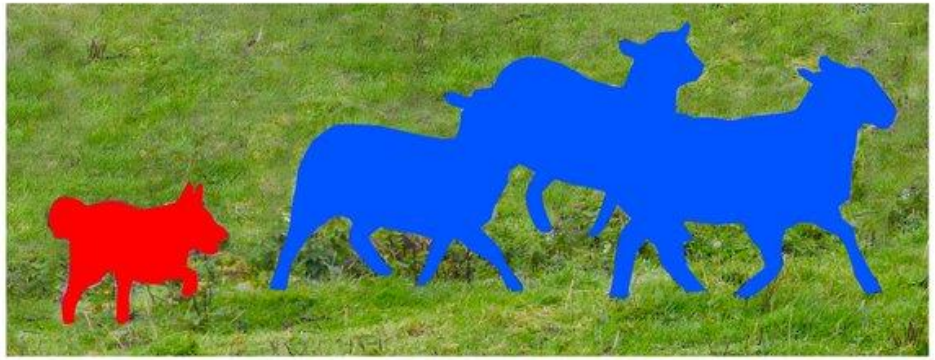
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Ways to Change Networks' Outputs

Faster R-CNN in 2015

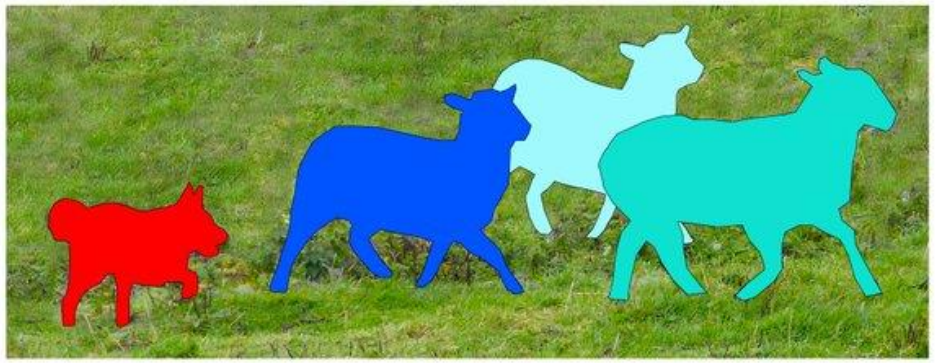


Object Detection



Semantic Segmentation

FCN in 2015



Instance Segmentation

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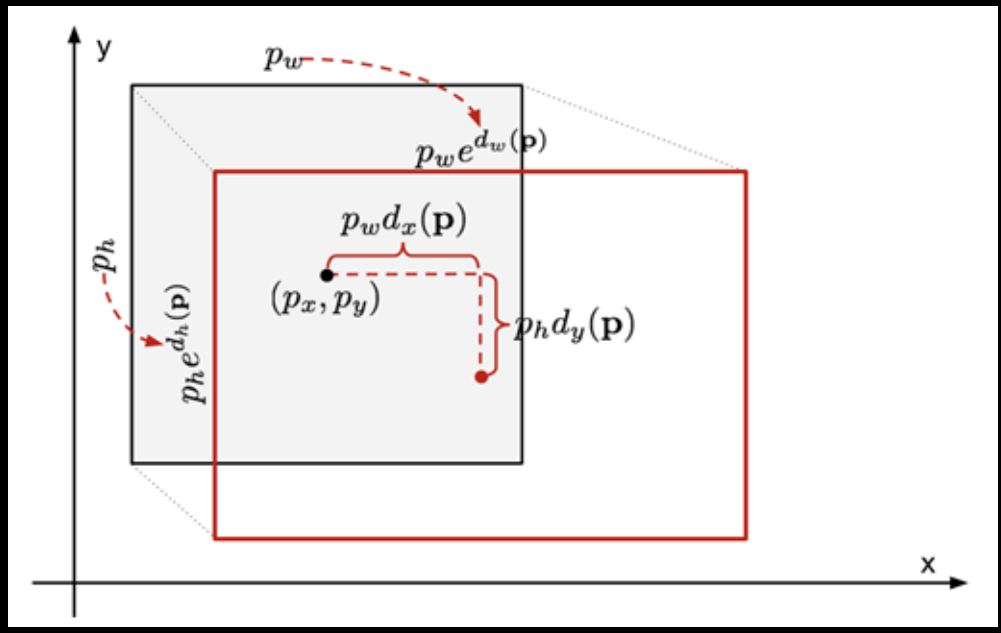
Architecture: Performs Two Tasks in Parallel

1. Bounding box regression: identifies adjustments to each object "region proposal" (p_x,p_y,p_h,p_w) to arrive to the correct location, by shifting its centroid (d_x, d_y) and scaling the width (d_w) and height (d_h)

- how many neurons are in the output layer?

- while predicted values are unbounded,

  training restricts them to a small range



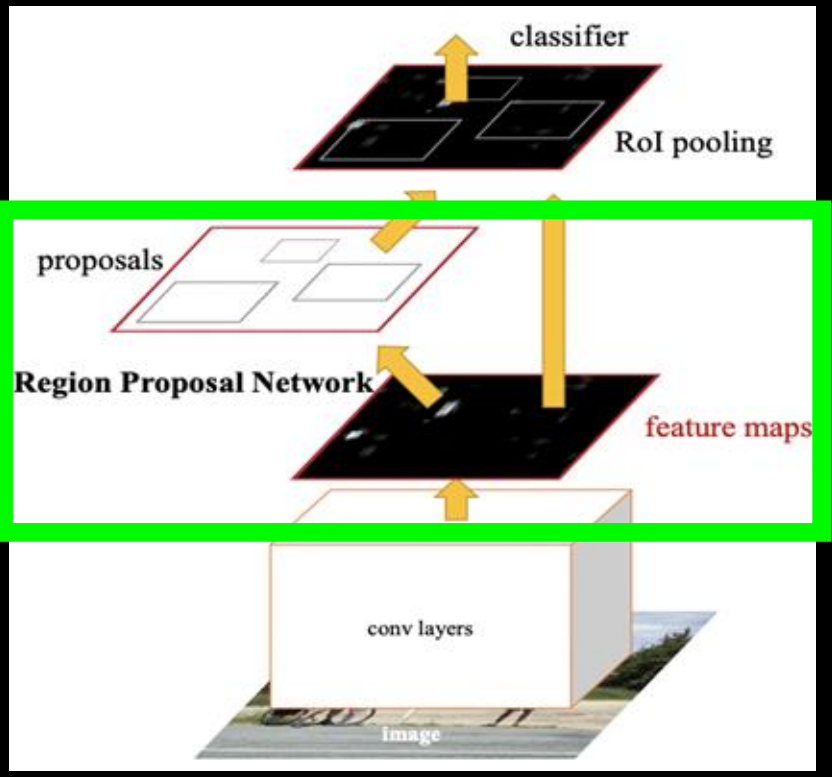
2. Classification: assigns a class to each region proposal; uses a fully connected layer that outputs class scores (after softmax)

https://lilianweng.github.io/lil-log/2017/12/31/object-recognition-for-dummies-part-3.html#bounding-box-regression

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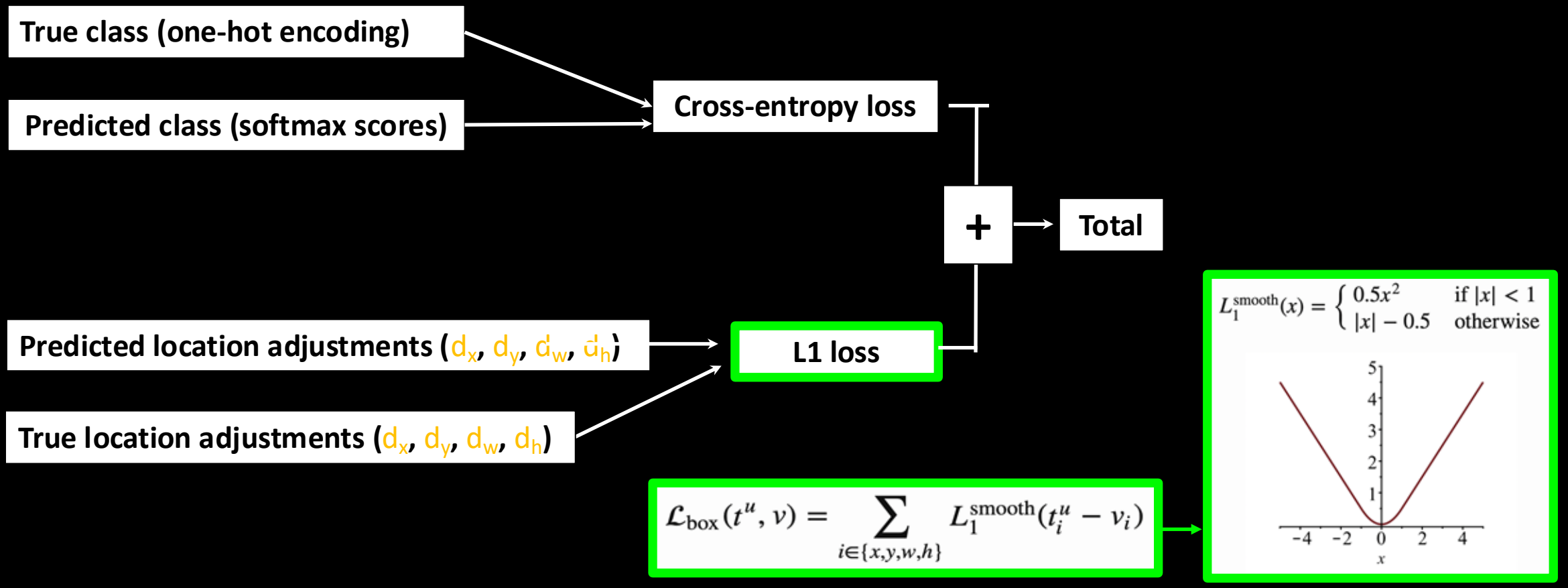
Idea: predicts manageable number of "object"-like regions rather than sliding window across entire image



Ren et al. Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks. Neurips 2015.

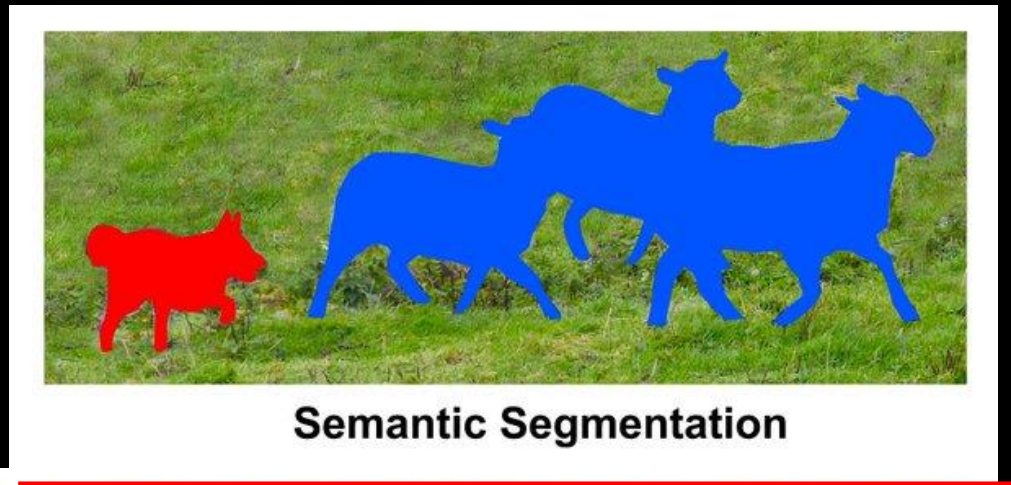
Objective Function: Multi-task Loss

Sum classification and localization losses (also used for region proposal network):



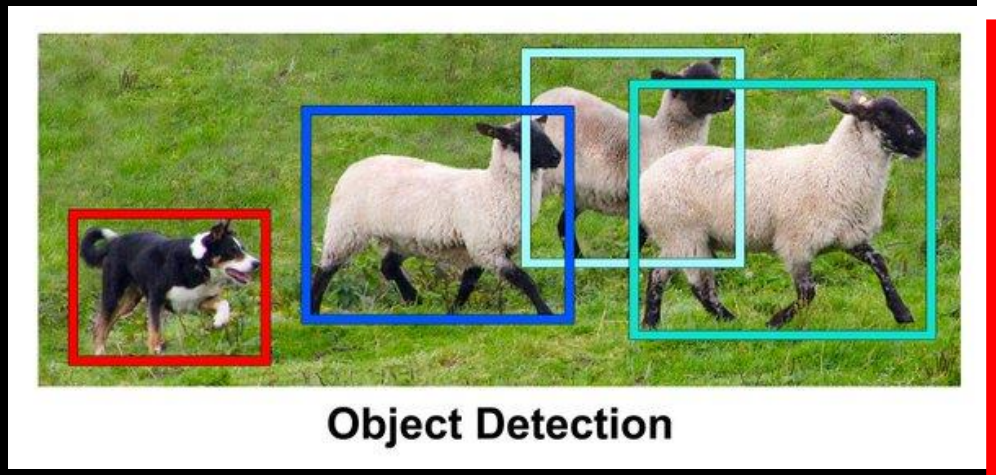
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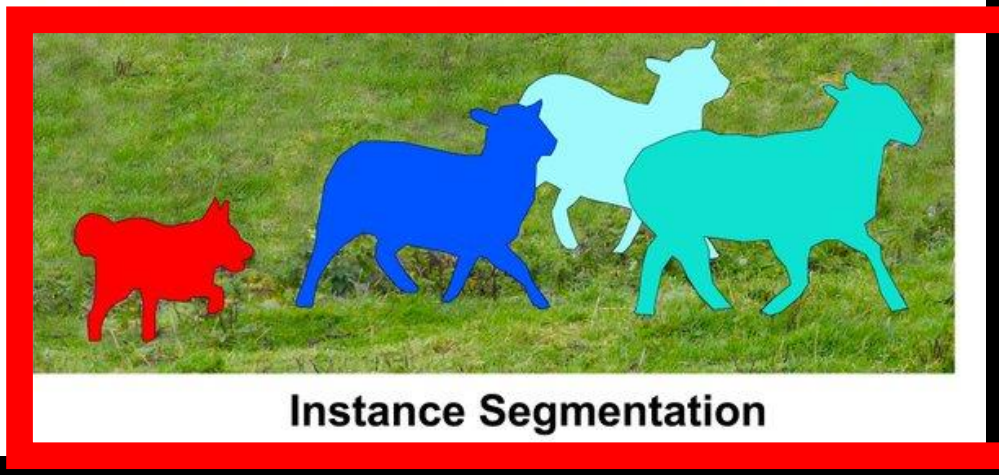
Ways to Change Networks' Outputs



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Faster R-CNN in 2015

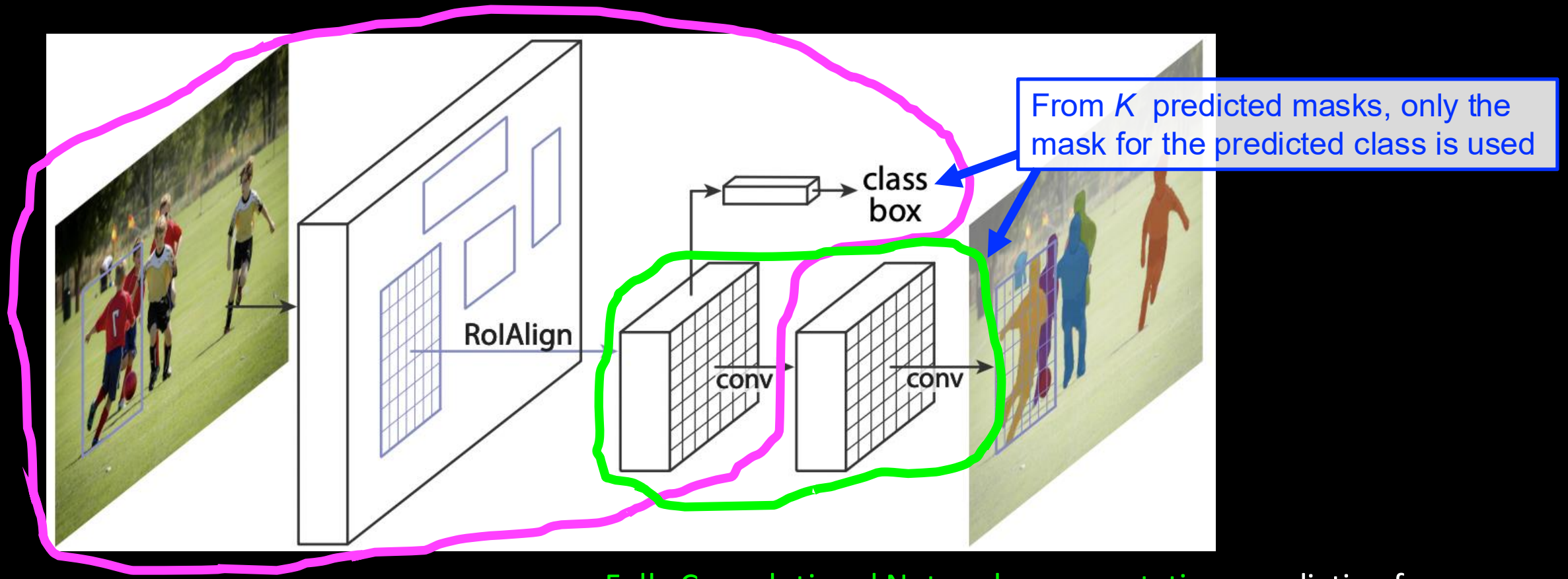




Mask R-CNN in 2015

https://ai-pool.com/d/could-you-explain-me-how-instance-segmentation-works

Architecture: Performs Three Tasks

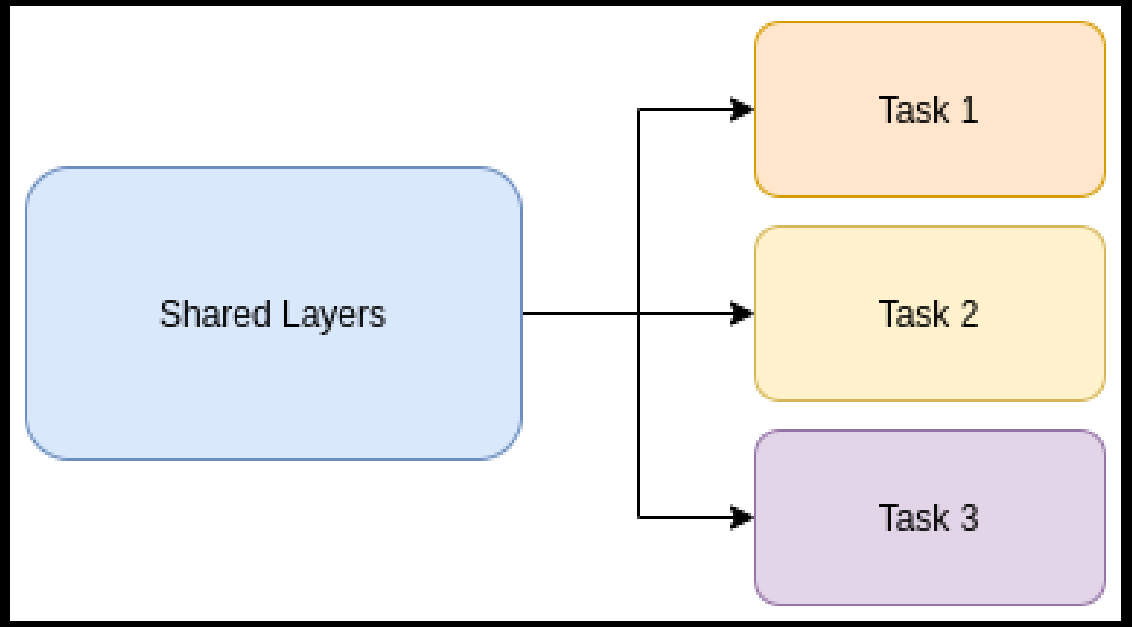


Faster R-CNN: object detection, predicting for each ROI regression values and class label

Fully Convolutional Network: segmentation, predicting for each ROI *K m* x *m* binary segmentation masks (i.e., for each of the K classes, applies per-pixel sigmoid followed by 0.5 thresholding to classify as foreground vs background)

He et al. Mask R-CNN. ICCV 2017

Objective Function: Multi-task Loss



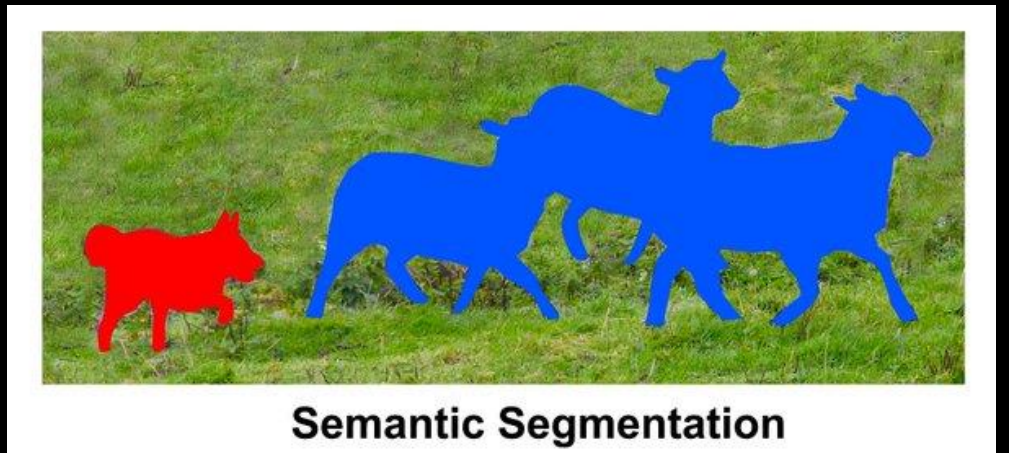
https://towardsdatascience.com/multi-task-learning-with-pytorch-and-fastai-6d10dc7ce855

$$L = L_{class} + L_{box} + L_{mask}$$

Same as Faster R-CNN

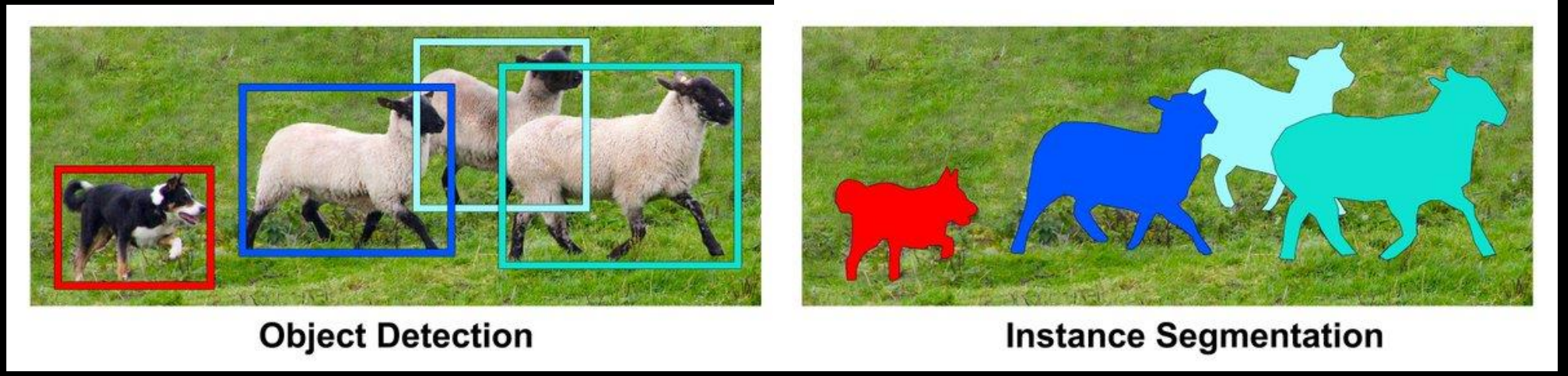
For each predicted mask of each ROI's target class, average binary cross-entropy loss is computed across all pixels

Summary: Different Tasks Have Different Network Outputs and Learning Objectives



FCN in 2015

Faster R-CNN in 2015



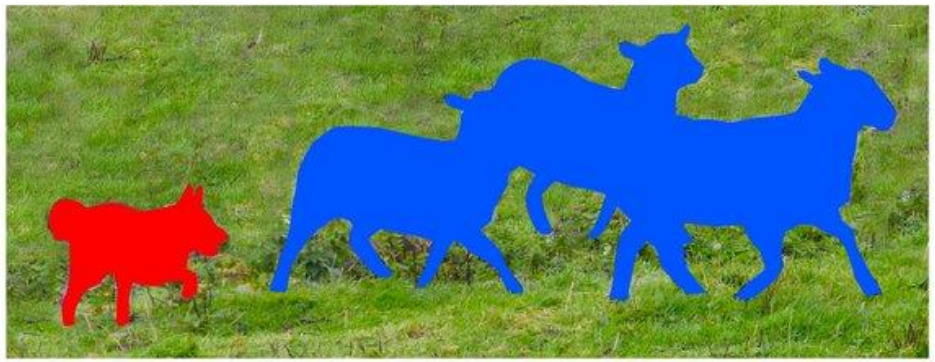
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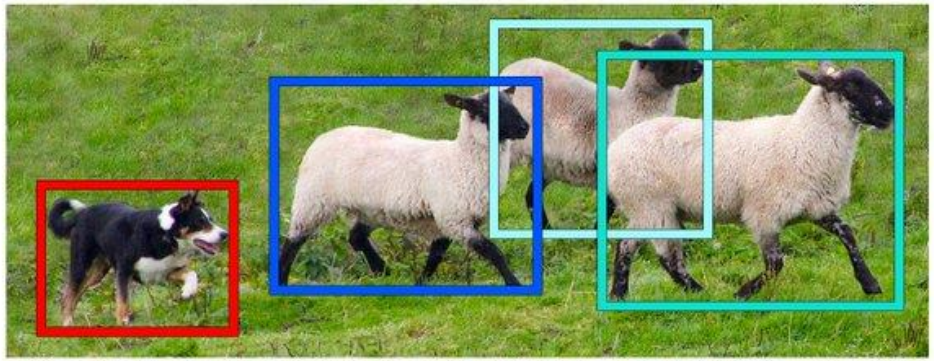
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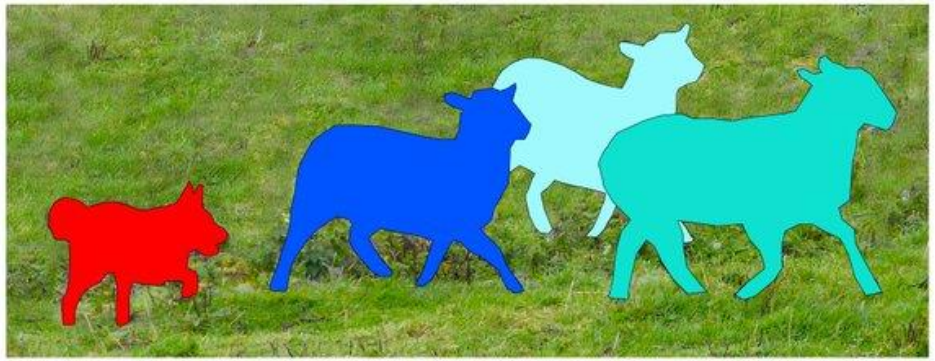
Ways to Support Arbitrary Input Dimensions



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Faster R-CNN in 2015

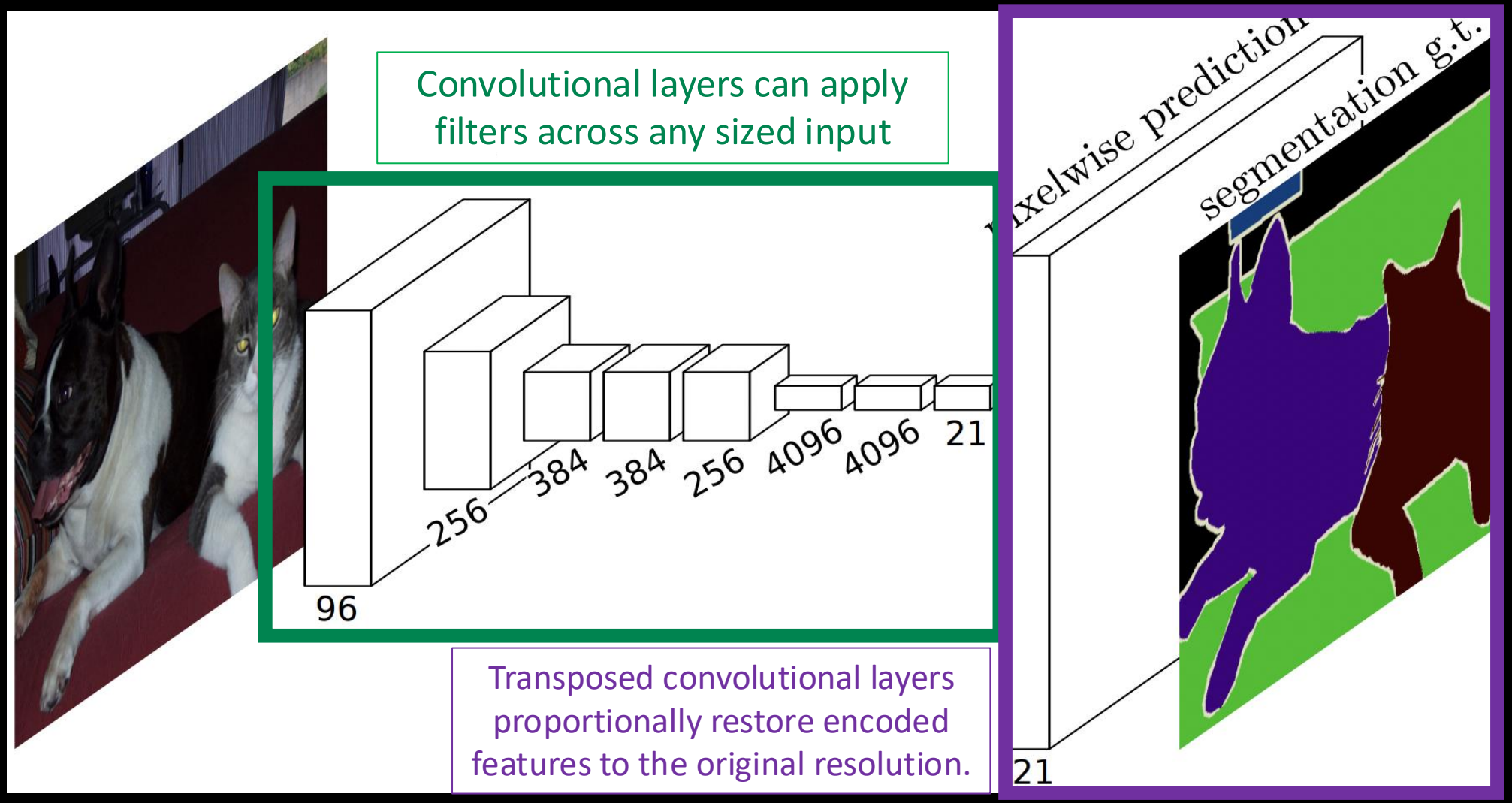




Mask R-CNN in 2015

https://ai-pool.com/d/could-you-explain-me-how-instance-segmentation-works

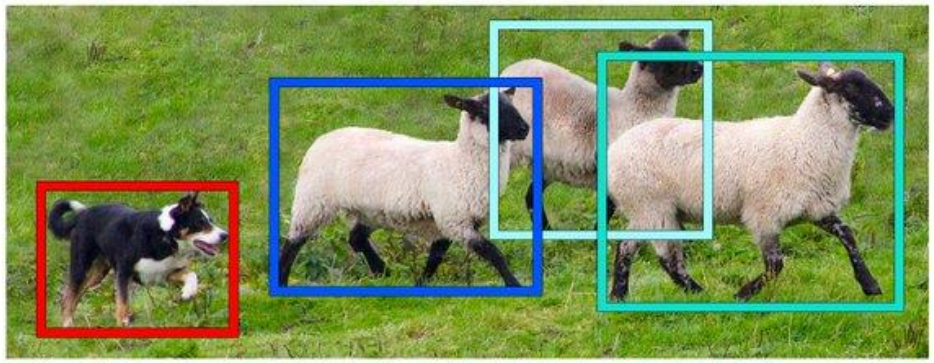
Architecture: All Convolutional Layers



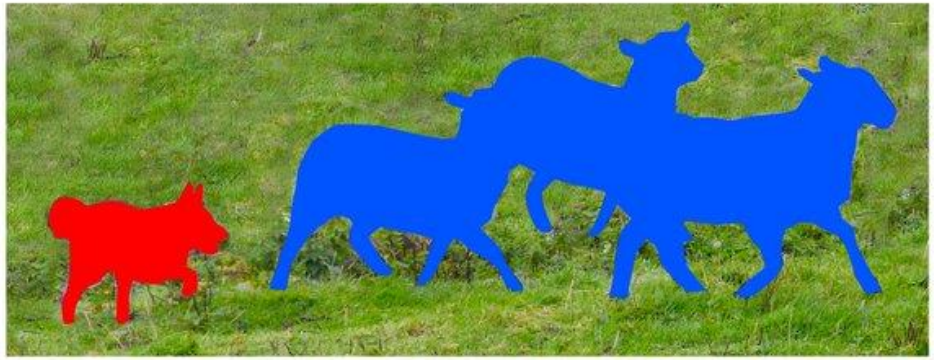
Long, Shelhamer, and Darrell. Fully Convolutional Networks for Semantic Segmentation. CVPR 2015.

Ways to Support Arbitrary Input Dimensions

Faster R-CNN in 2015

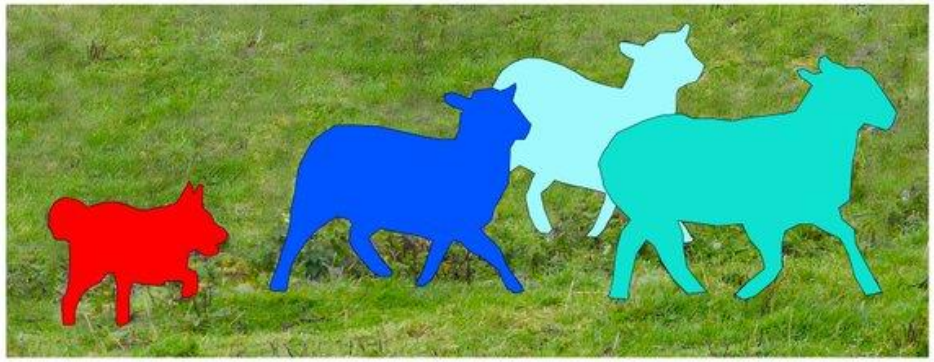


Object Detection



Semantic Segmentation

FCN in 2015

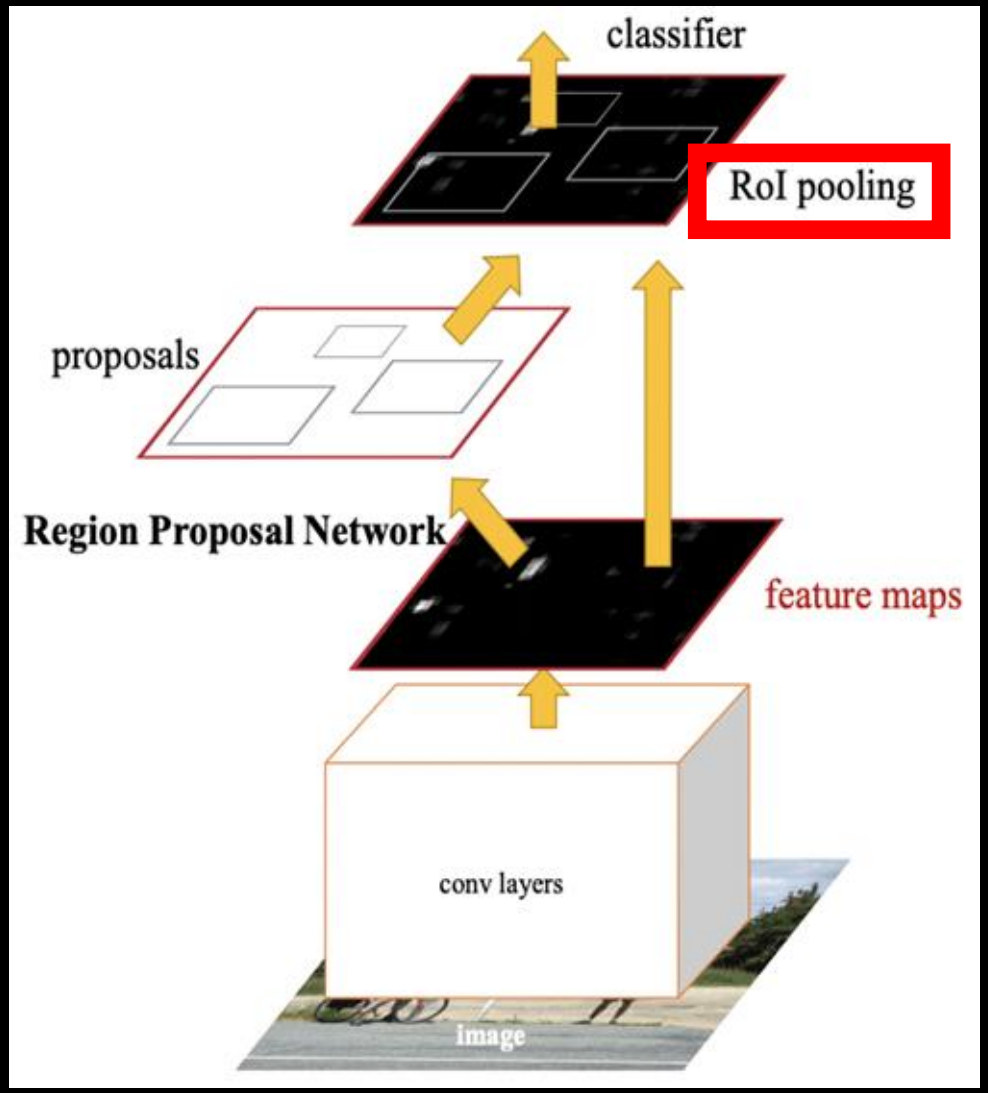


Instance Segmentation

Mask R-CNN in 2015

https://ai-pool.com/d/could-you-explain-me-how-instance-segmentation-works

ROI Pooling: Quantization + Pooling



Pools any sized input into a target size;

e.g., [6, 4] -> [3, 3]

- What discrete integer would convert 6 columns into 3

  6/3 = 2

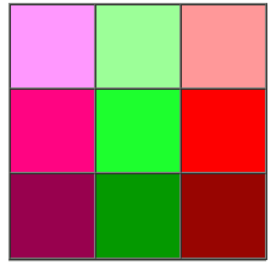
- What discrete integer would convert 4 rows into 3

  4/3 = 1.3 -> 1

4x6 RoI

0.1	0.2	0.3	0.4	0.5	0.6
1	0.7	0.2	0.6	0.1	0.9
0.9	0.8	0.7	0.3	0.5	0.2

3x3 RoI Pooling



https://erdem.pl/2020/02/understanding-region-of-interest-ro-i-pooling

Ren et al. Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks. Neurips 2015.

Today's Topics

- Motivation: when vision asks not just "what", but also "where"
- Redesigning neural networks to predict densely, rather than just classify
- Freeing dense prediction networks from fixed input sizes
- Pivotal insights: pretrained features + design tricks to boost spatial precision
- Programming tutorial

Key Challenge (1 of 2): Train Large Capacity Models

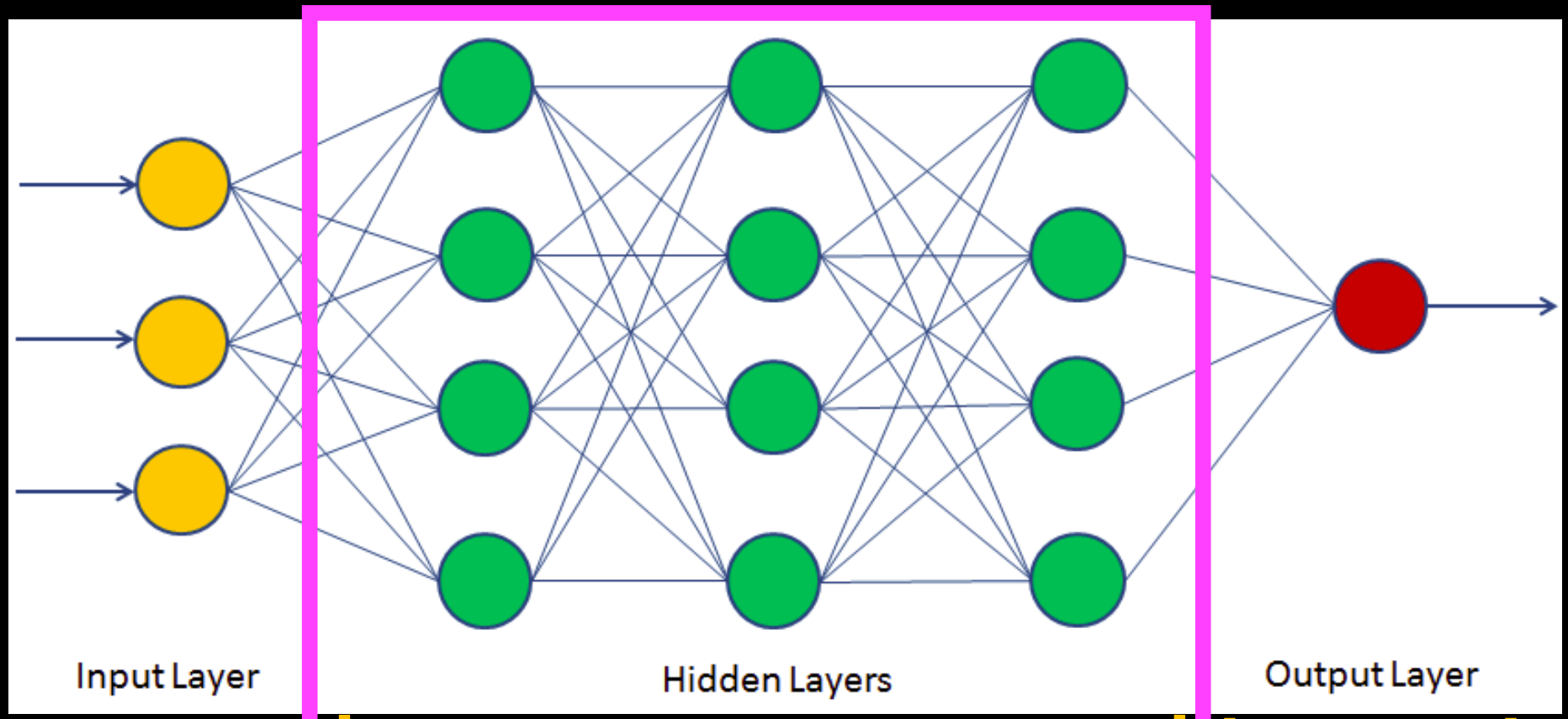
- VERY challenging to collect large-scale annotated datasets
- How long do you think it would take to draw a:
  - bounding box? (a) 5-15 seconds, (b) 16-45 seconds, (c) 46-75 seconds
  - segmentation?



Avg time from 101 annotators: 7 seconds per bounding box 54 seconds per segmentation

Jain and Grauman. Predicting Sufficient Annotation Strength for Interactive Foreground Segmentation. ICCV 2013

Key Idea: Transfer Learning



Use a pretrained backbone to bypass needing a large-scale training dataset

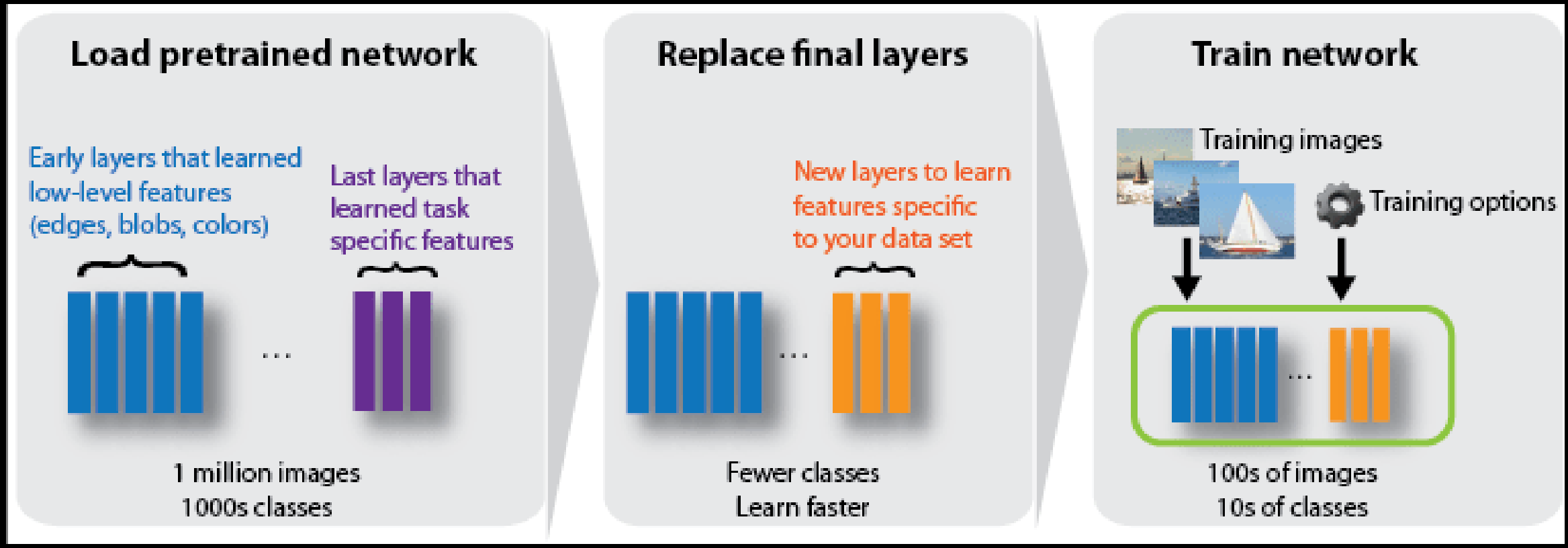
How to extract features characterizing data

How to predict

https://www.datacamp.com/community/tutorials/neural-network-models-r

Key Idea: Transfer Learning

Use pretrained network as a starting point to train for a different dataset and/or task; e.g.,



https://www.mathworks.com/help/deeplearning/ug/transfer-learning-using-alexnet.html

Key Idea: Transfer Learning

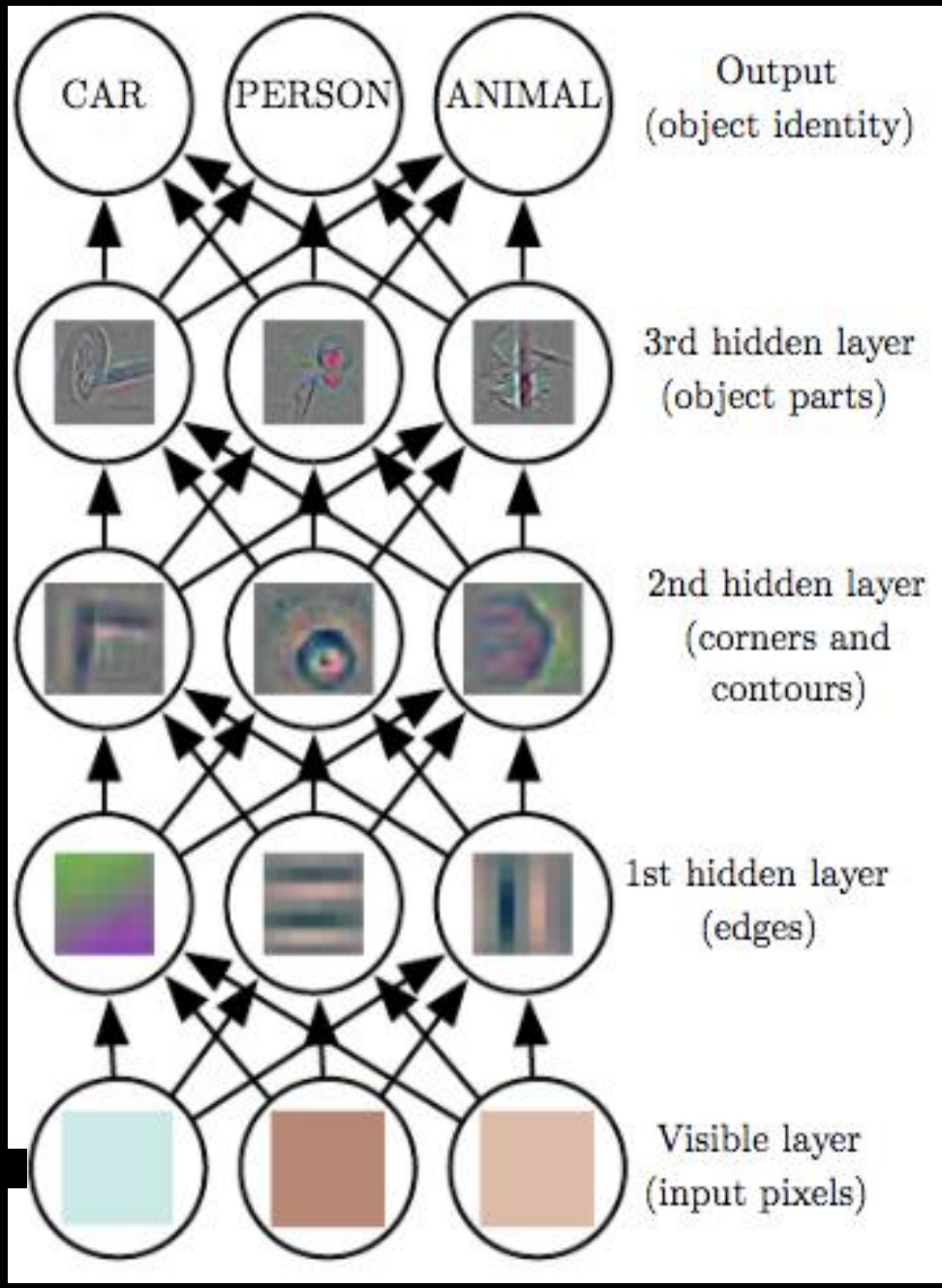
Critical Questions:

1. Which pretrained network?
2. Which layers from the network?
3. Freeze vs train transferred layers (latter is called “fine-tuning”)

High-level features

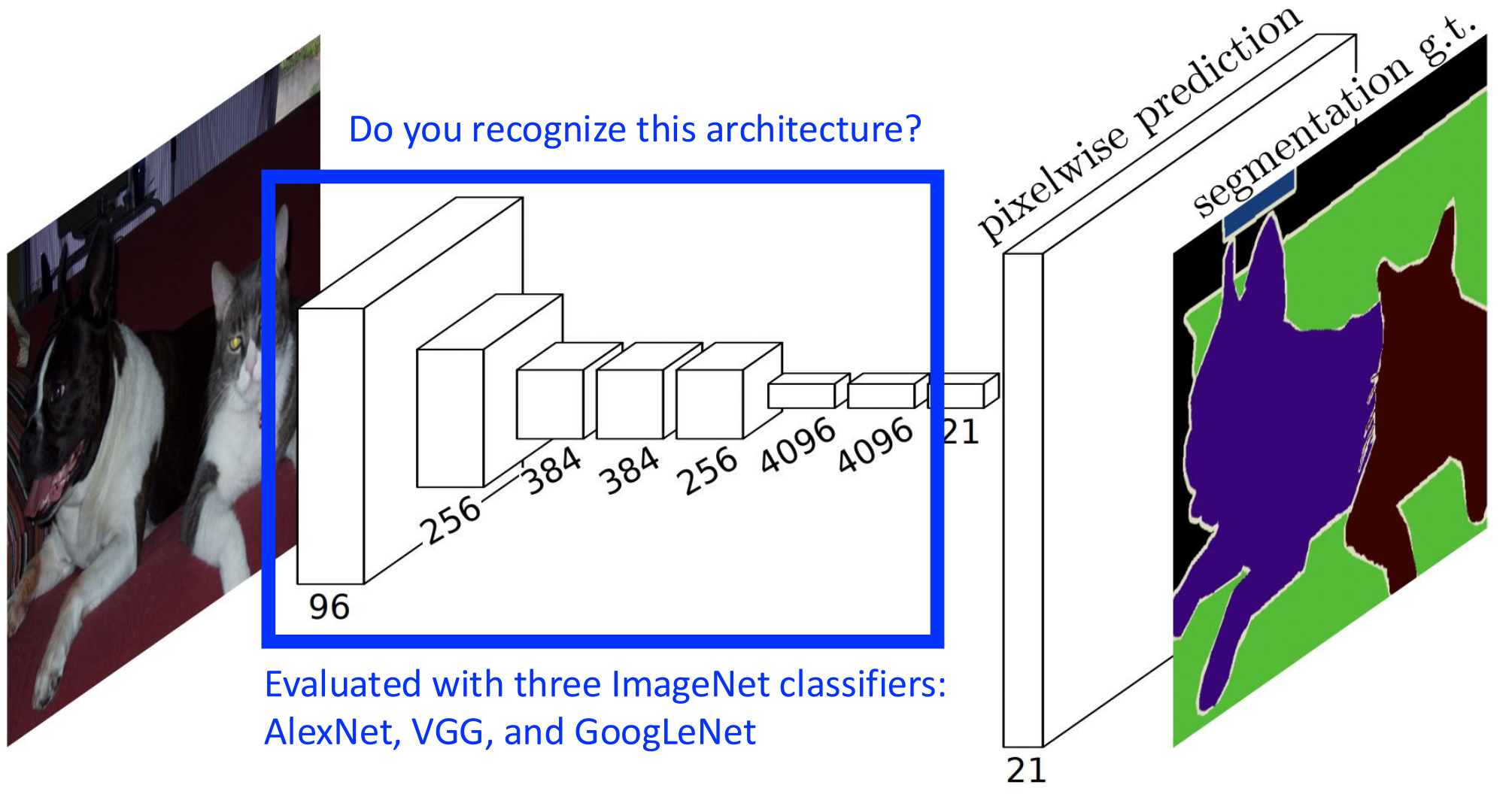
Mid-level features

Low-level features



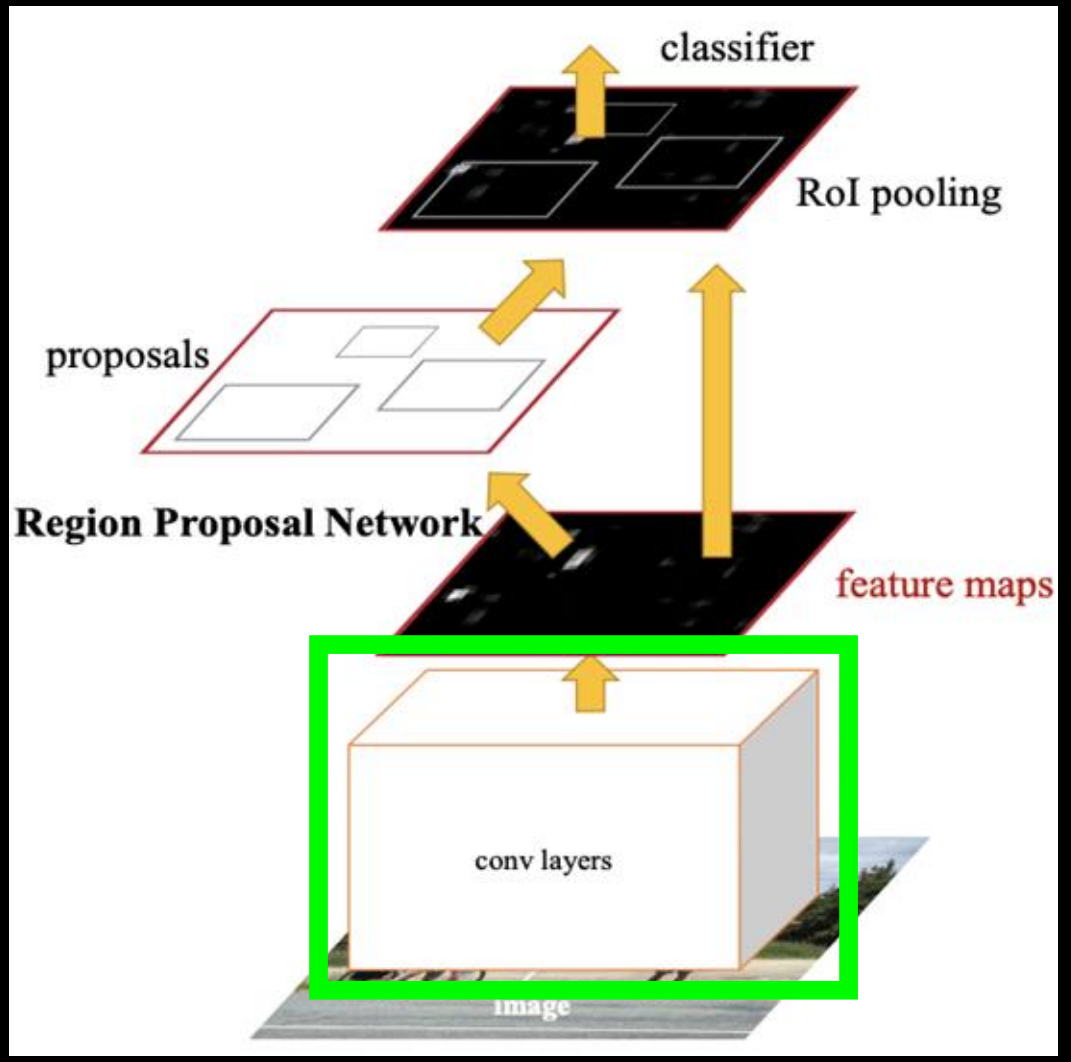
Deep Learning, Ian Goodfellow, Yoshua Bengio, and Aaron Courville

FCN (Semantic Segmentation) Backbone



Long, Shelhamer, and Darrell. Fully Convolutional Networks for Semantic Segmentation. CVPR 2015.

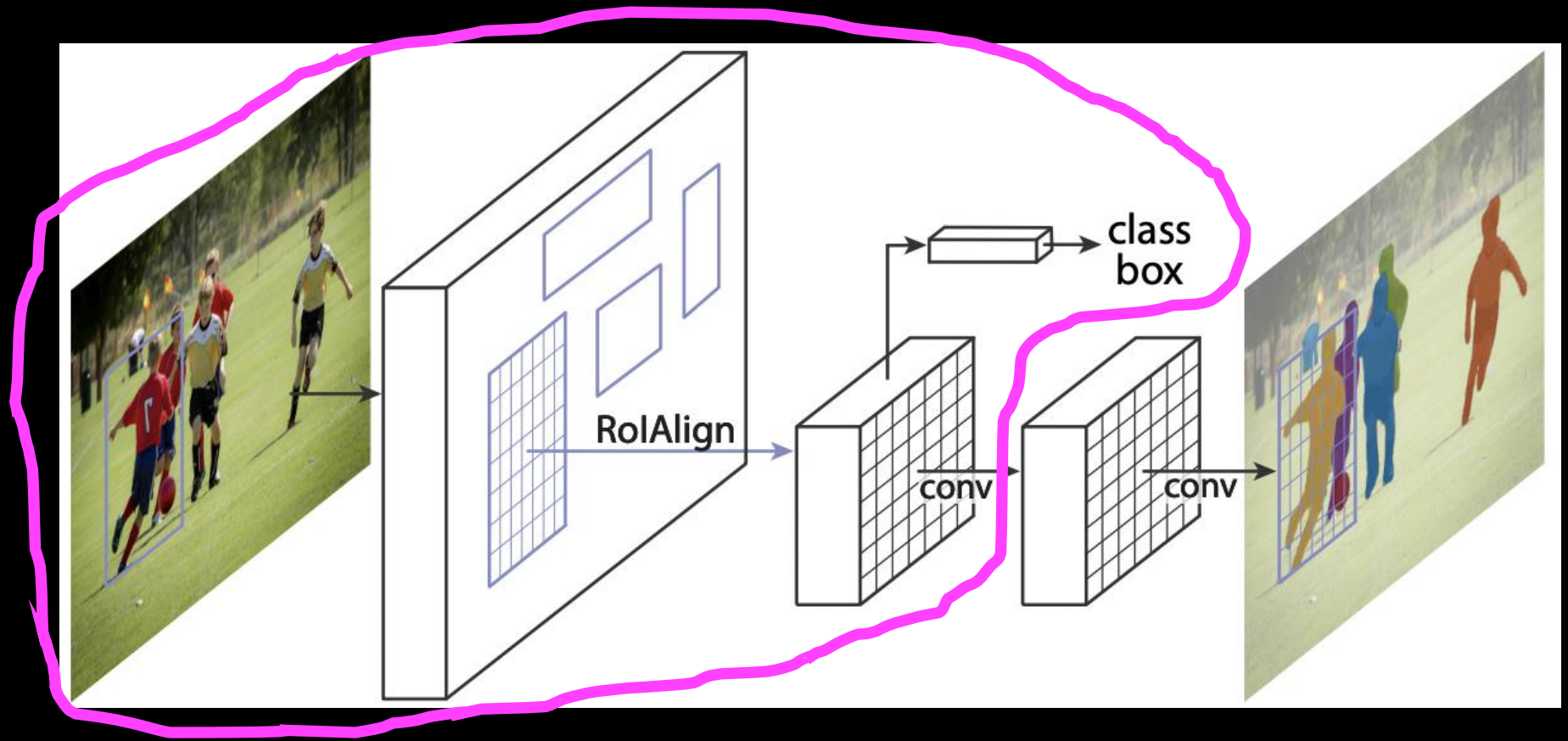
Faster R-CNN (Object Detection) Backbone



Evaluated with two ImageNet classifiers:
Zeiler and Fergus model and VGG-16

Ren et al. Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks. Neurips 2015.

Mask R-CNN (Instance Segmentation) Backbone

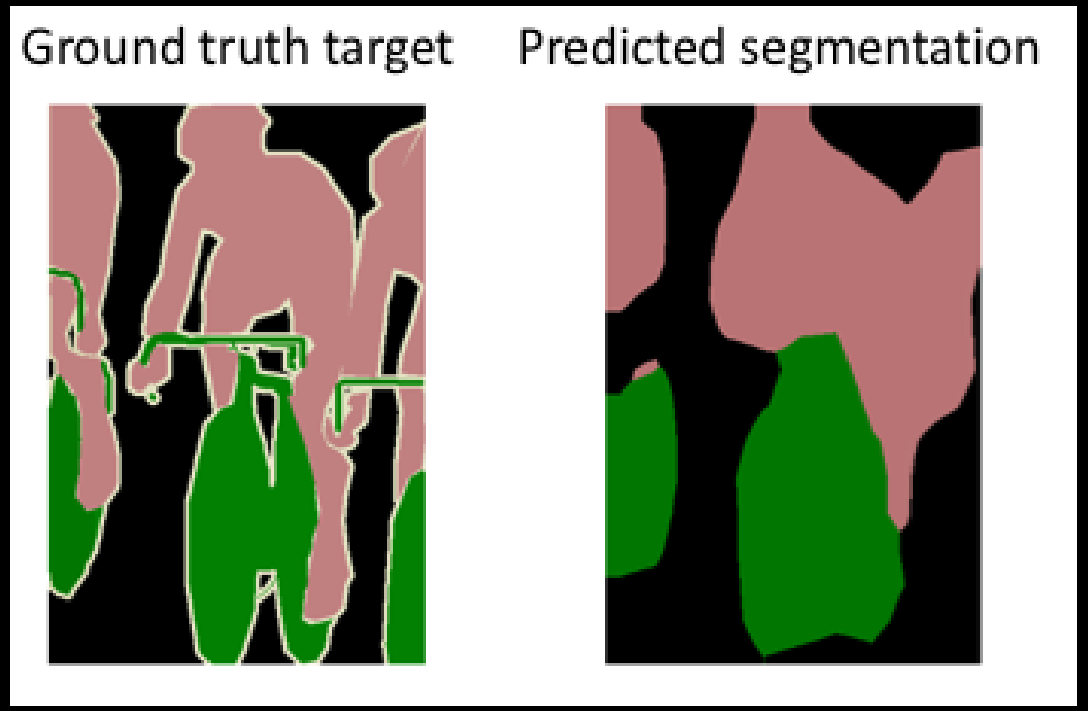


Embedded in the Faster R-CNN module,
three ImageNet classifiers evaluated:
ResNet, ResNeXt, & Feature Pyramid Network

He et al. Mask R-CNN. ICCV 2017

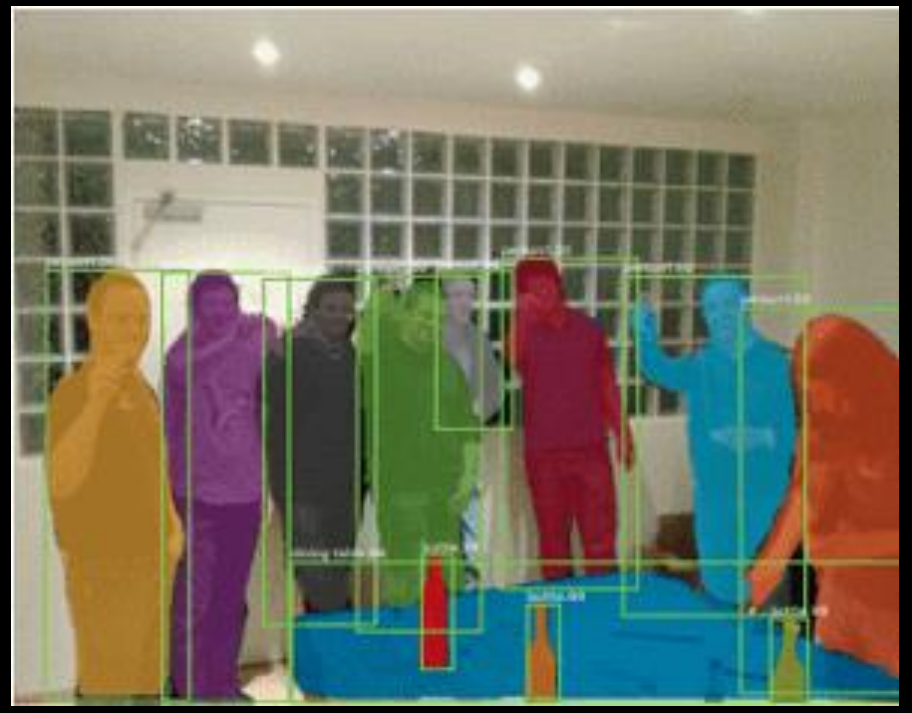
Key Challenge (2 of 2): Moving Beyond Coarse Accuracy for Segmentation

FCN Results



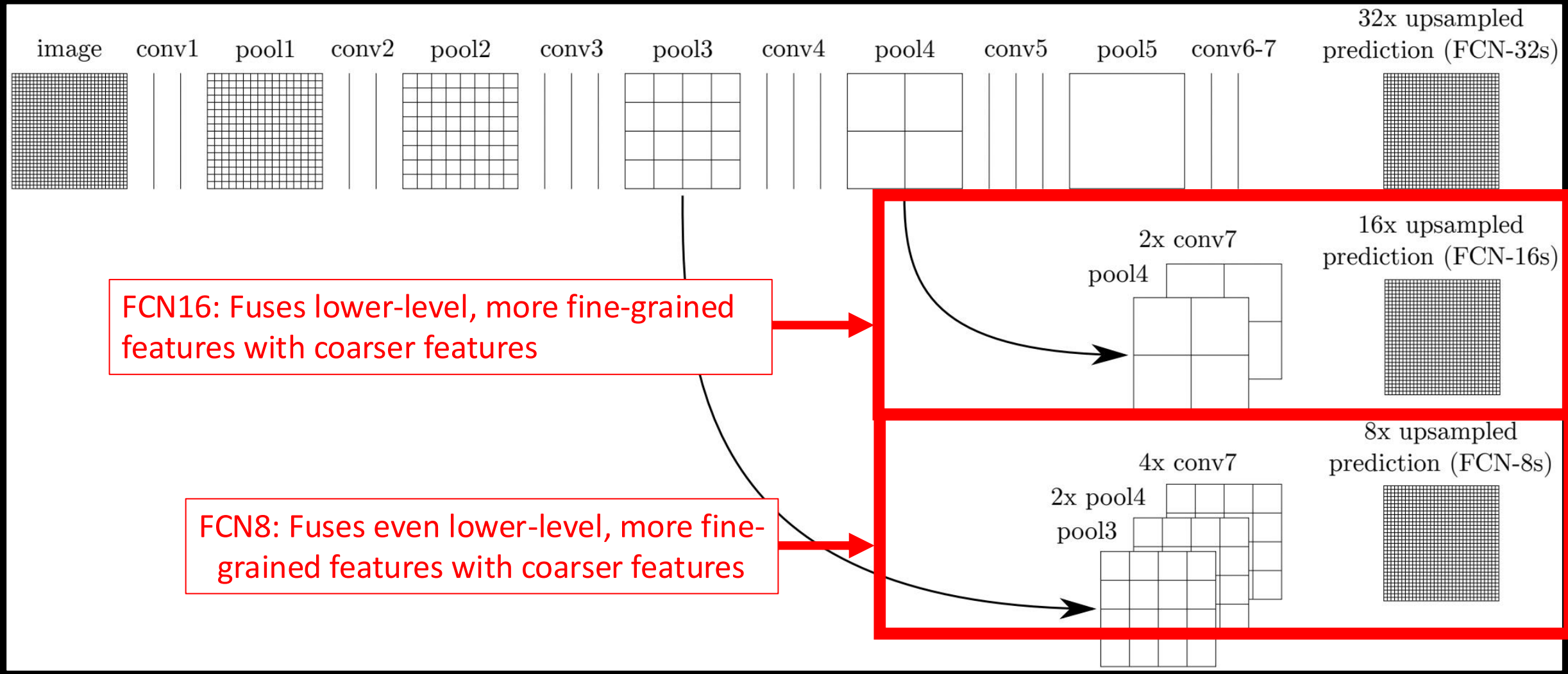
Long, Shelhamer, and Darrell. Fully Convolutional Networks for Semantic Segmentation. CVPR 2015.

Mask R-CNN Results (BB vs Segmentation)



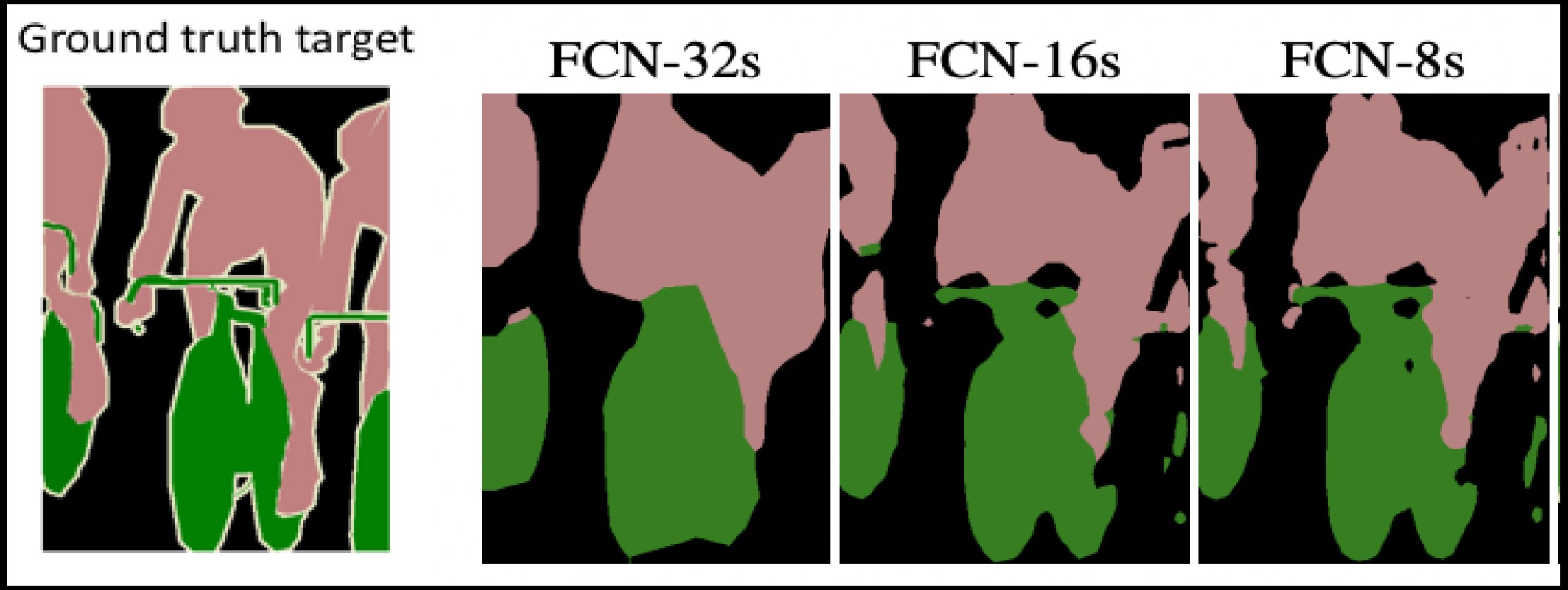
He et al. Mask R-CNN. ICCV 2017

FCN: Add Skip Connections to Fuse Coarse Features with Finer-Grained Features



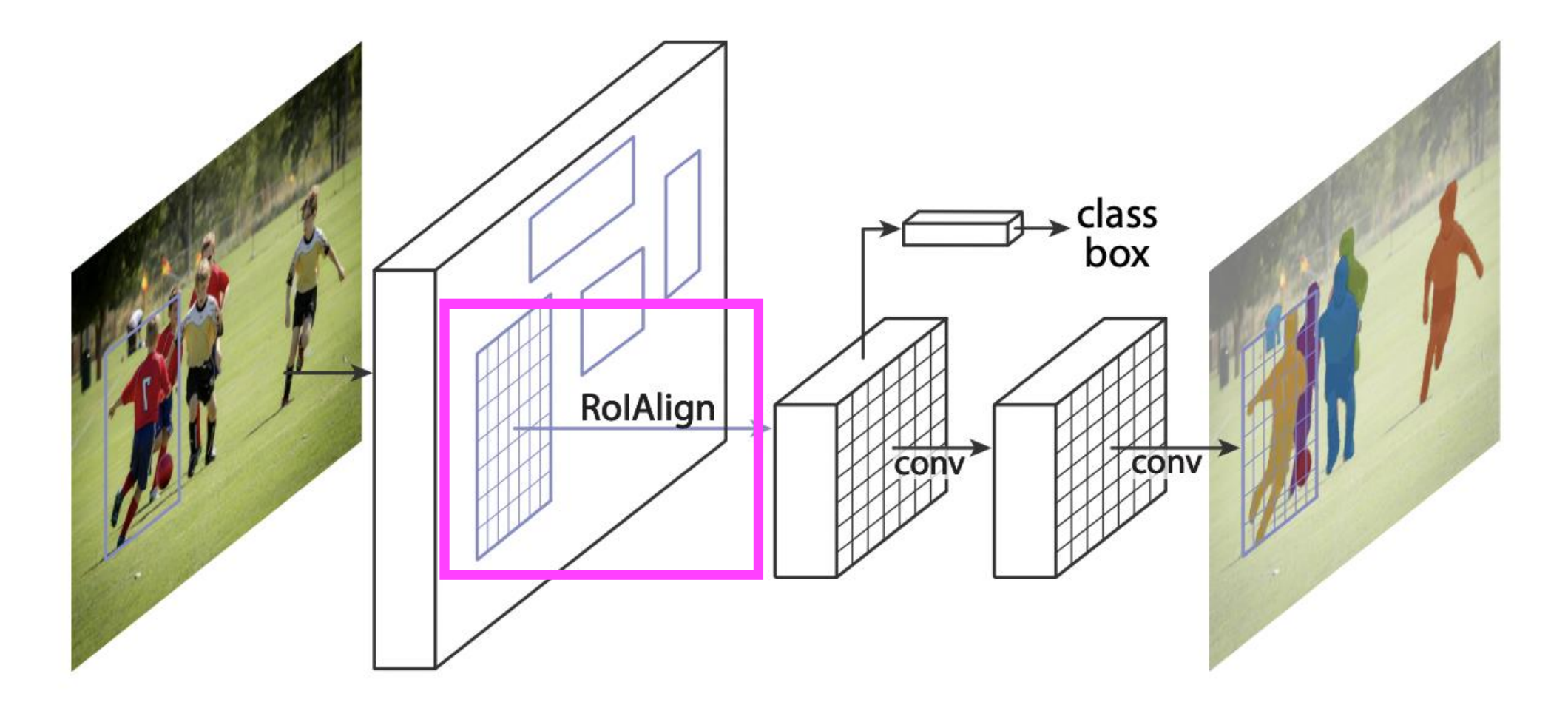
Long, Shelhamer, and Darrell. Fully Convolutional Networks for Semantic Segmentation. CVPR 2015.

FCN: Add Skip Connections to Fuse Coarse Features with Finer-Grained Features



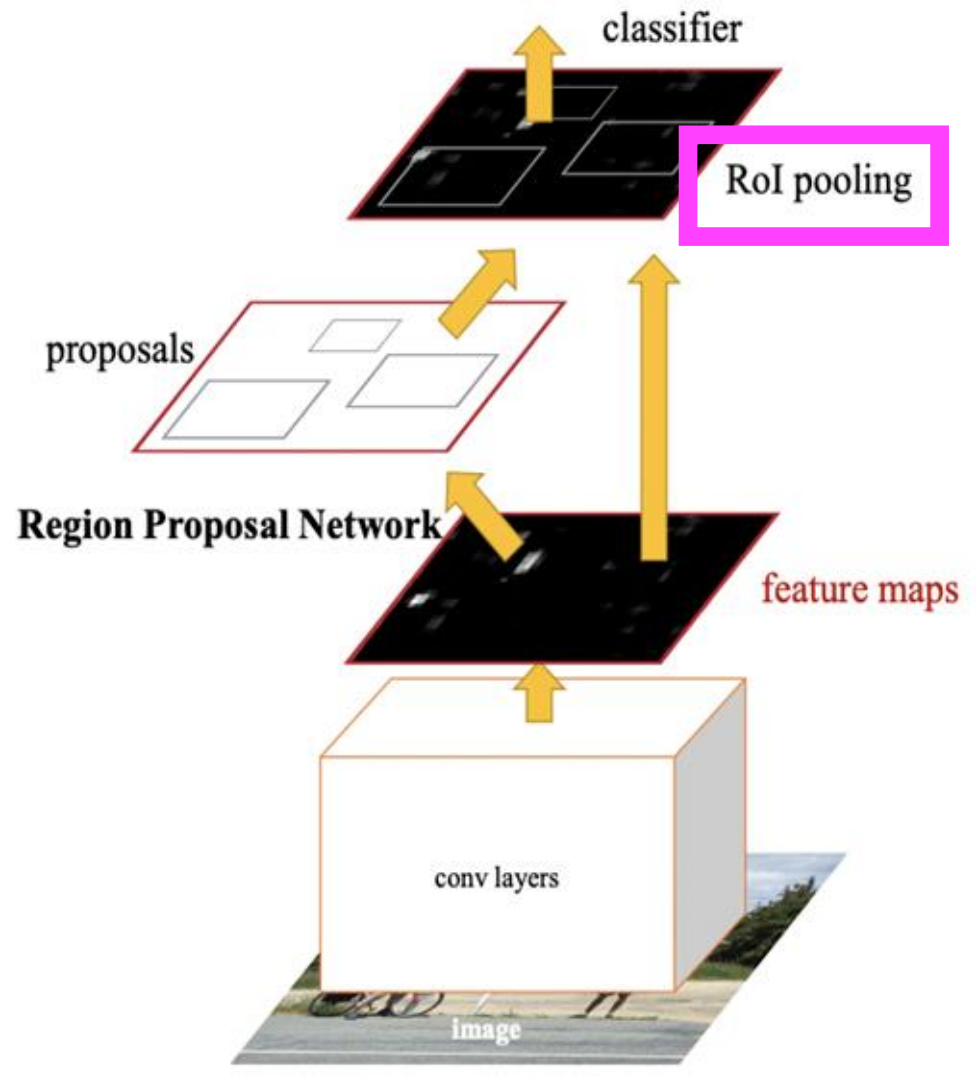
Long, Shelhamer, and Darrell. Fully Convolutional Networks for Semantic Segmentation. CVPR 2015.

Mask R-CNN: Replace ROI-Pool with ROI-Align



He et al. Mask R-CNN. ICCV 2017

ROI-Pool Limitation

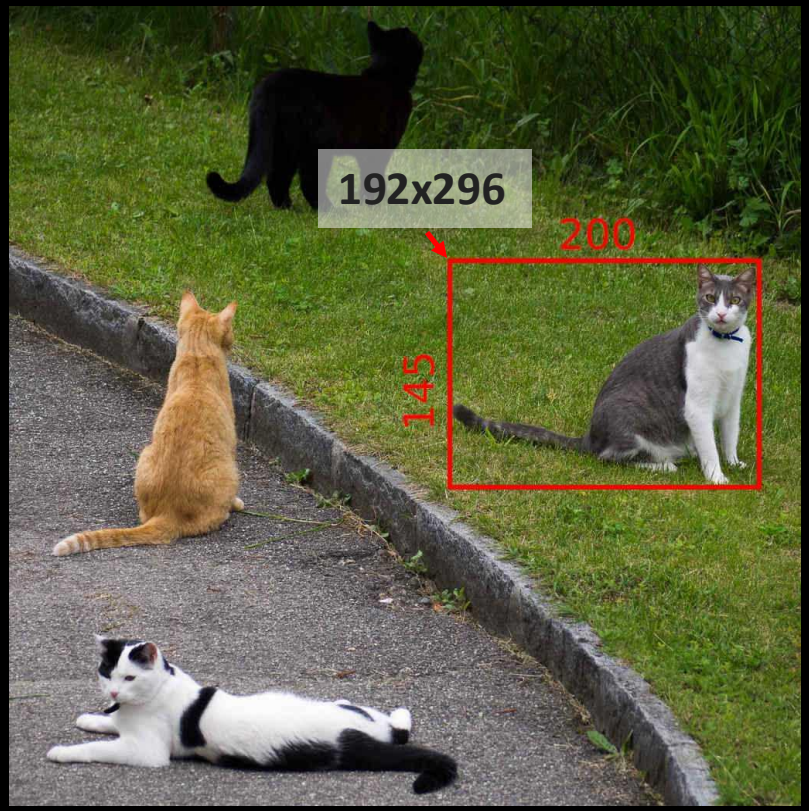


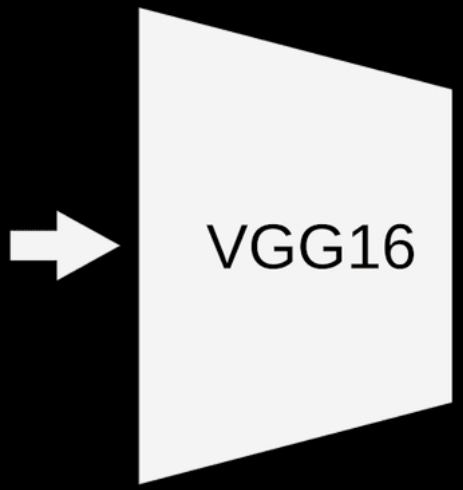
Ren Shaoqing Ren et al. Neurips 2015

Problem 1: quantizing ROIs in downsized feature map

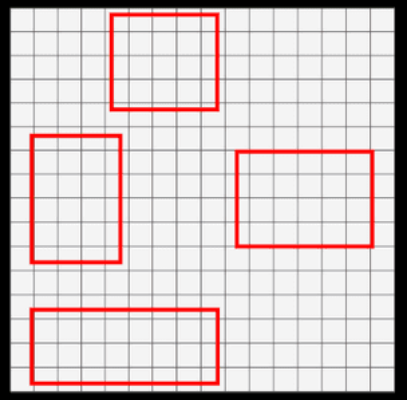
ROI-Pool Limitation

What are the values for the region in the original image in the downsampled feature map?





(1/32 of original size)

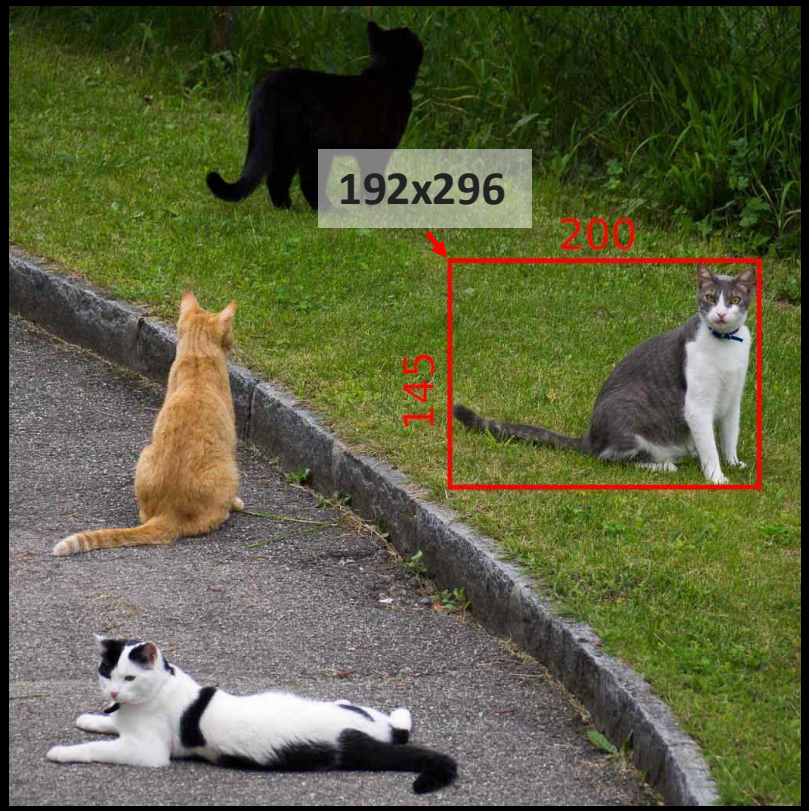


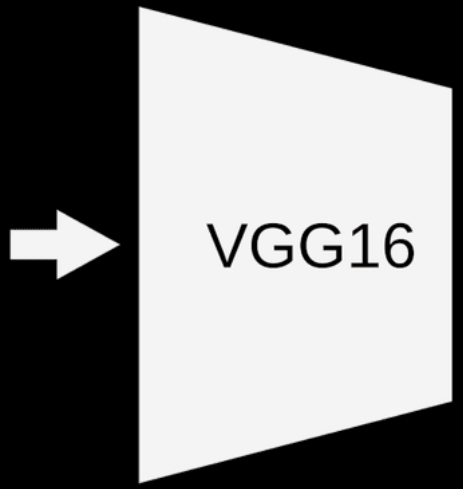
Width: ?
Height: ?
Upper-left X: ?
Upper-left Y: ?

https://erdem.pl/2020/02/understanding-region-of-interest-ro-i-pooling

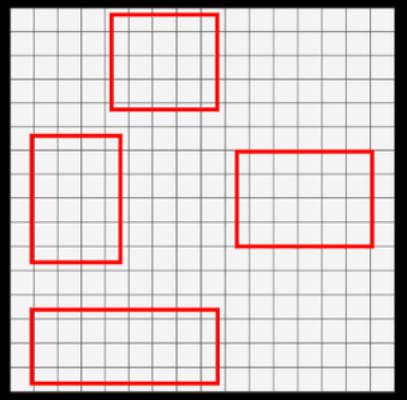
ROI-Pool Limitation

What are the values for the region in the original image in the downsampled feature map?





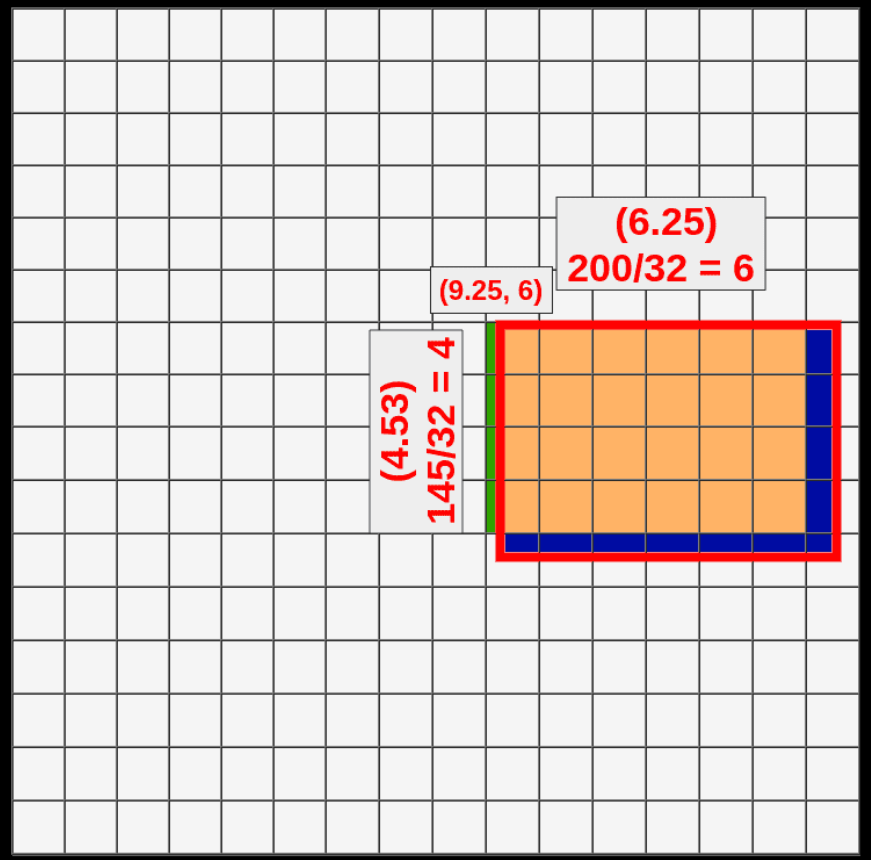
(1/32 of original size)



Width: 200/32 = 6.25
Height: 145/32 = ~4.53
Upper-left X: 192/32 = 9.25
Upper-left Y: 145/32 = 6

https://erdem.pl/2020/02/understanding-region-of-interest-ro-i-pooling

ROI-Pool Limitation



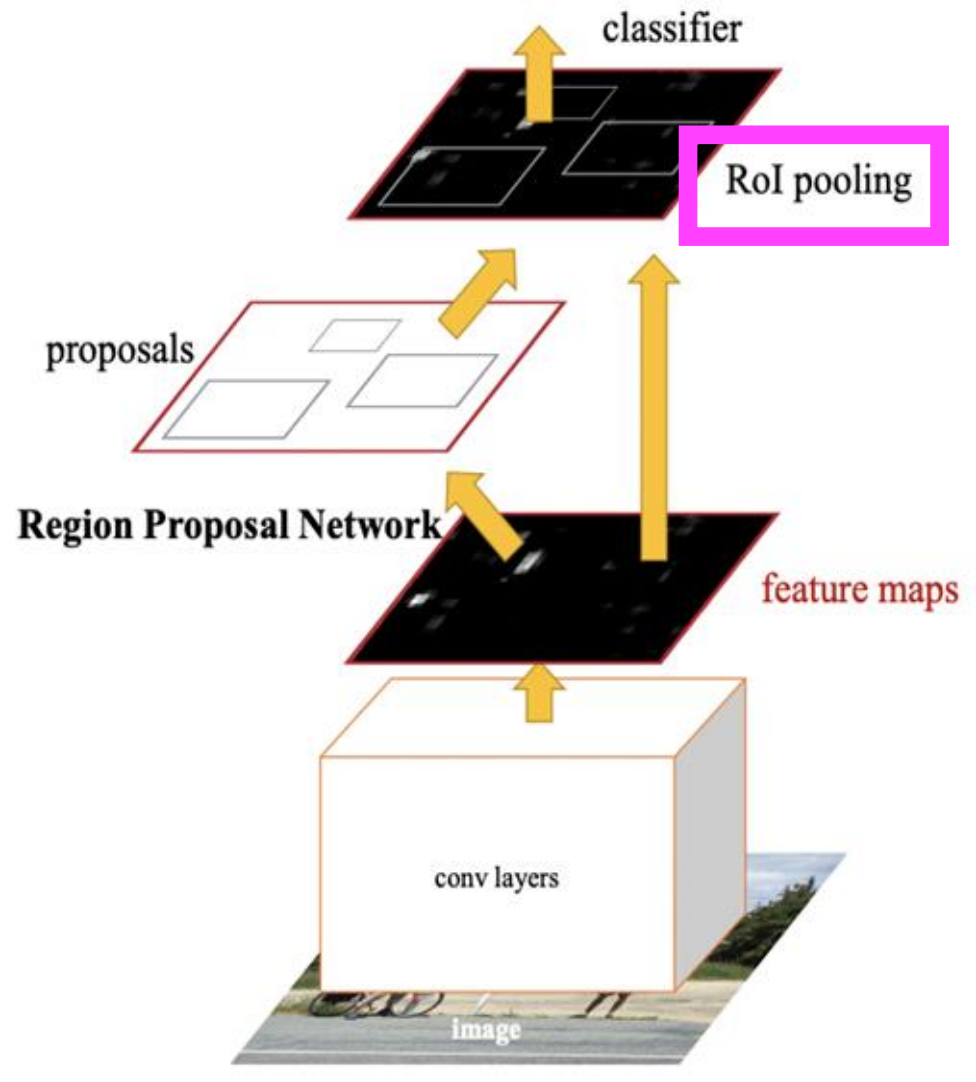
Original region on feature map

Quantized variant:values rounded down to discrete set of integers to match the grid
- Original information preserved
- Information added
- Information lost

Quantization changes the information utilized from the original image, losing information about the object and adding extra image context (recall, the original image is orders of magnitude larger than the feature map!)

https://erdem.pl/2020/02/understanding-region-of-interest-ro-i-pooling

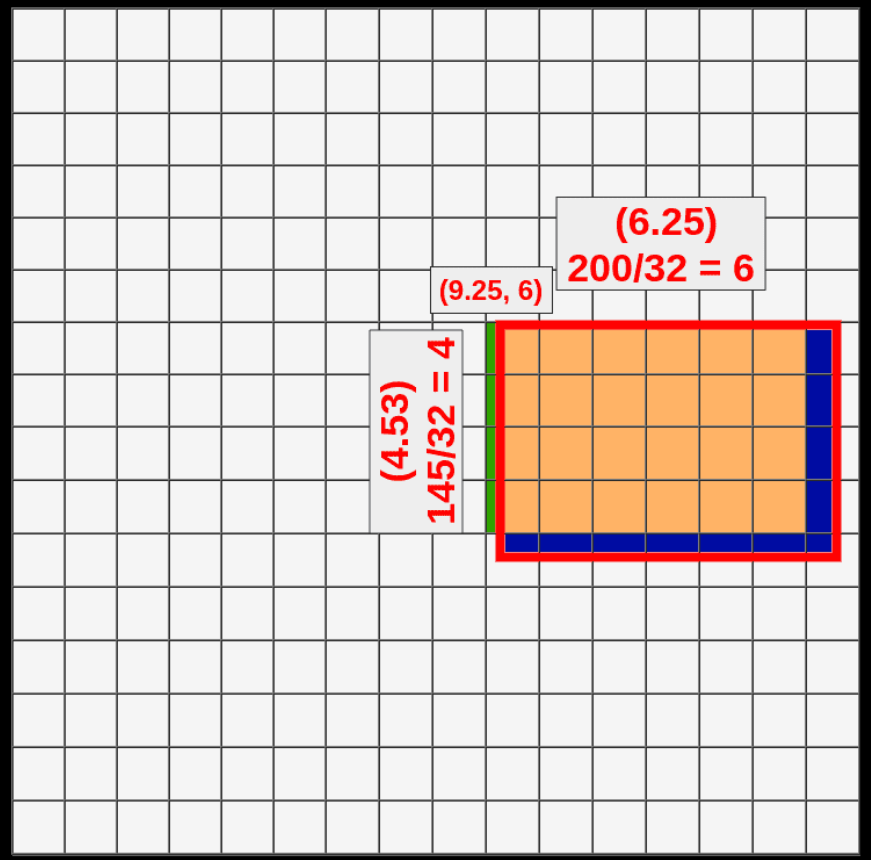
ROI-Pool Limitation



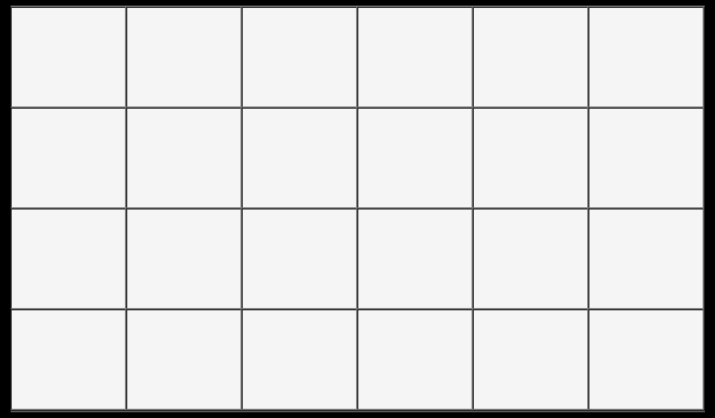
Ren Shaoqing Ren et al. Neurips 2015

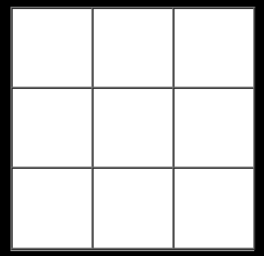
Problem 2: Quantization when pooling ROIs of various sizes to target size required for fully connected layer

ROI-Pool Limitation



e.g., convert quantized 4x6 region into a 3x3 feature



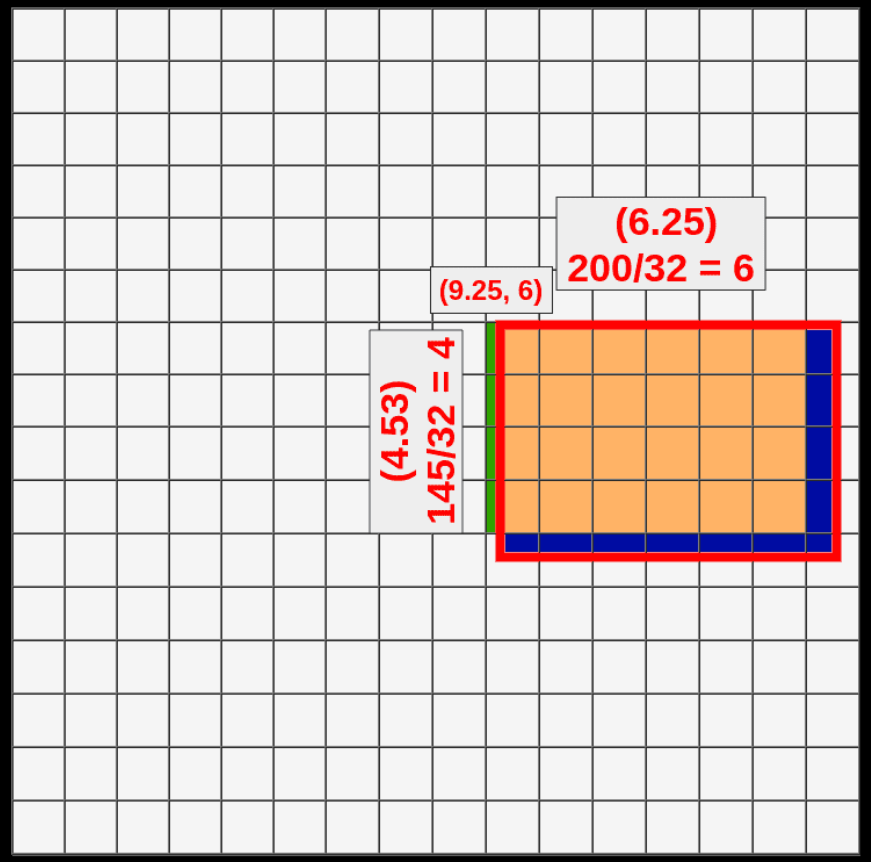


Quantized approach: identify discrete integers for pooling to result in the target size

e.g., 4/3 = 1.3 -> 1 and 6/3 = 2

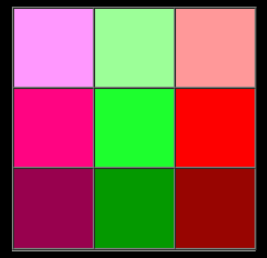
https://erdem.pl/2020/02/understanding-region-of-interest-ro-i-pooling

ROI-Pool Limitation



e.g., convert quantized 4x6 region into a 3x3 feature

0.1	0.2	0.3	0.4	0.5	0.6
1	0.7	0.2	0.6	0.1	0.9
0.9	0.8	0.7	0.3	0.5	0.2

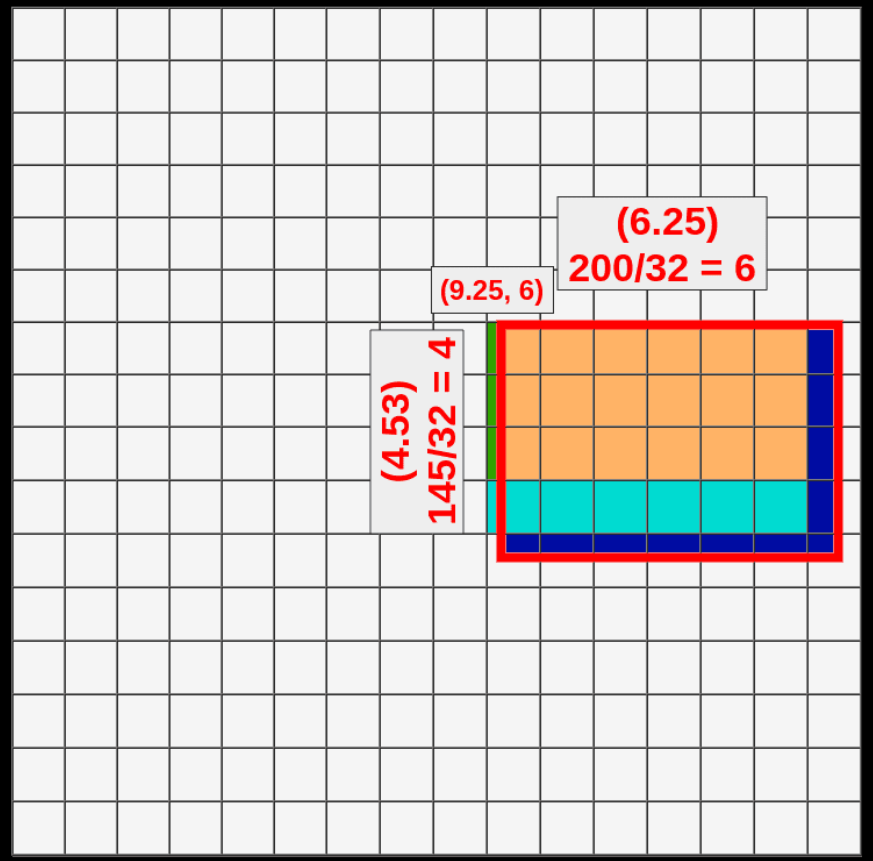


Quantized approach: identify discrete integers for pooling to result in the target size

e.g., 1x2 vector using max pooling

https://erdem.pl/2020/02/understanding-region-of-interest-ro-i-pooling

ROI-Pool Limitation



e.g., convert quantized 4x6 region into a 3x3 feature

0.1	0.2	0.3	0.4	0.5	0.6
1	0.7	0.2	0.6	0.1	0.9
0.9	0.8	0.7	0.3	0.5	0.2
0.2	0.5	1	0.7	0.1	0.1

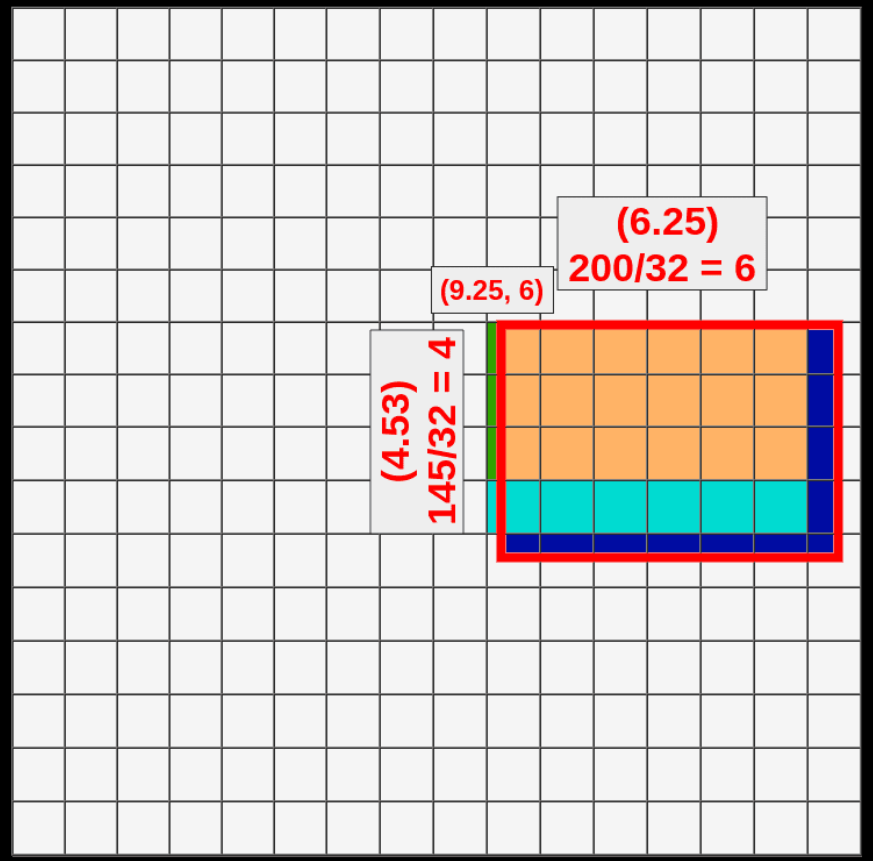
Again, quantization discards information about the object from the original image (recall, the original image is orders of magnitude larger than the feature map!)

Quantized approach: identify discrete integers for pooling to result in the target size

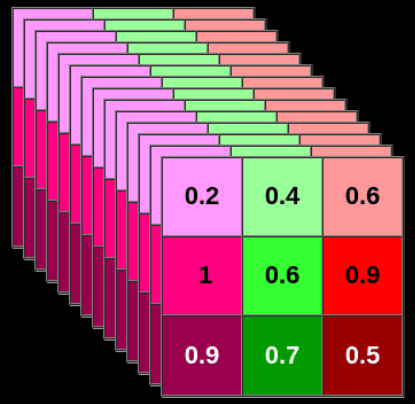
e.g., 1x2 vector using max pooling

https://erdem.pl/2020/02/understanding-region-of-interest-ro-i-pooling

ROI-Pool Limitation



e.g., convert quantized 4x6 region into a 3x3 feature



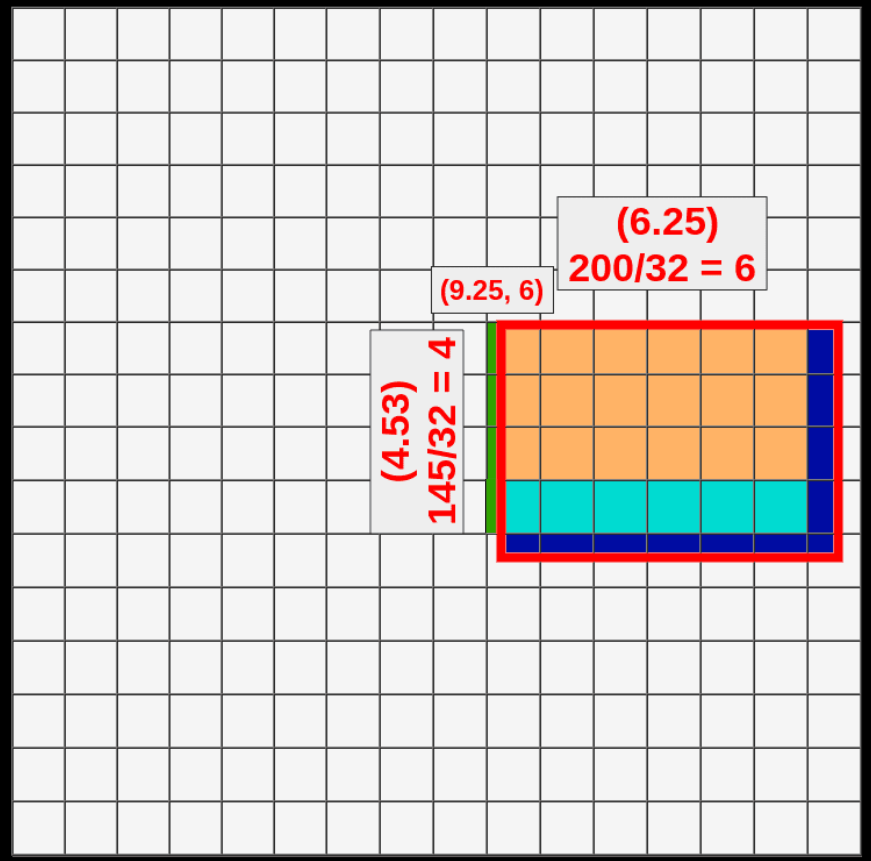
Information is lost for *all* channels for *every* region proposal (each of which is used to predict a class and bounding box)!

Quantized approach: identify discrete integers for pooling to result in the target size

e.g., 1x2 vector using max pooling

https://erdem.pl/2020/02/understanding-region-of-interest-ro-i-pooling

ROI-Pool Limitations Summary

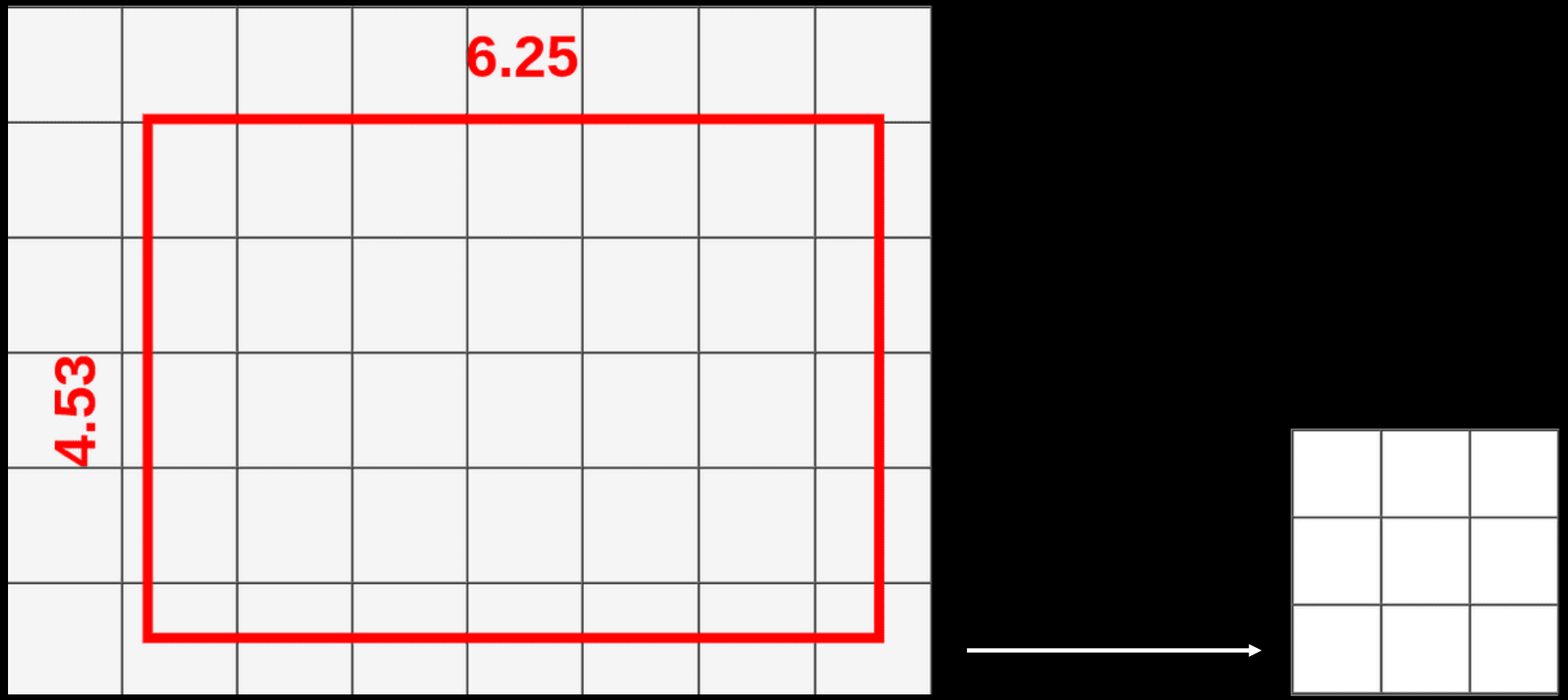


Original region on feature map

Quantization changes the information utilized from the original image, losing information about the object and adding extra image context (recall, the original image is orders of magnitude larger than the feature map!)

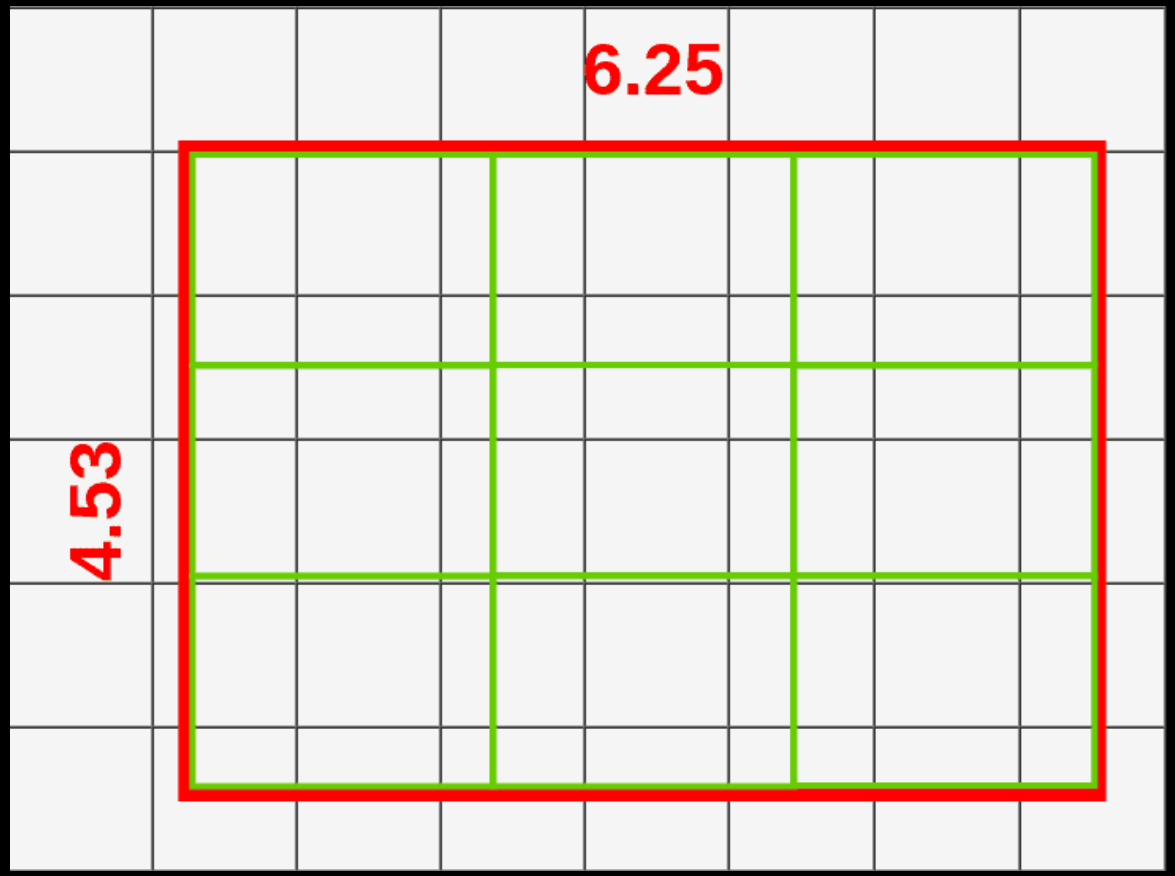
https://erdem.pl/2020/02/understanding-region-of-interest-ro-i-pooling

ROIAlign: Pooling *Without* Quantization



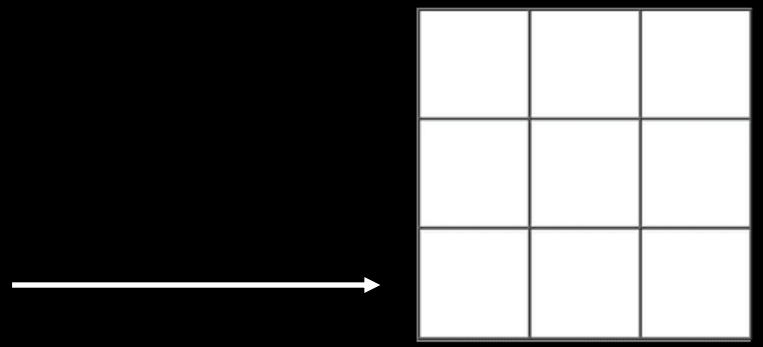
https://erdem.pl/2020/02/understanding-region-of-interest-ro-i-pooling

ROIAlign: Pooling *Without* Quantization



Divide region into 9 equal sized boxes; what is the size of each box?

- 6.25/3 x 4.53/3 = 2.08 x 1.51



https://erdem.pl/2020/02/understanding-region-of-interest-ro-i-pooling

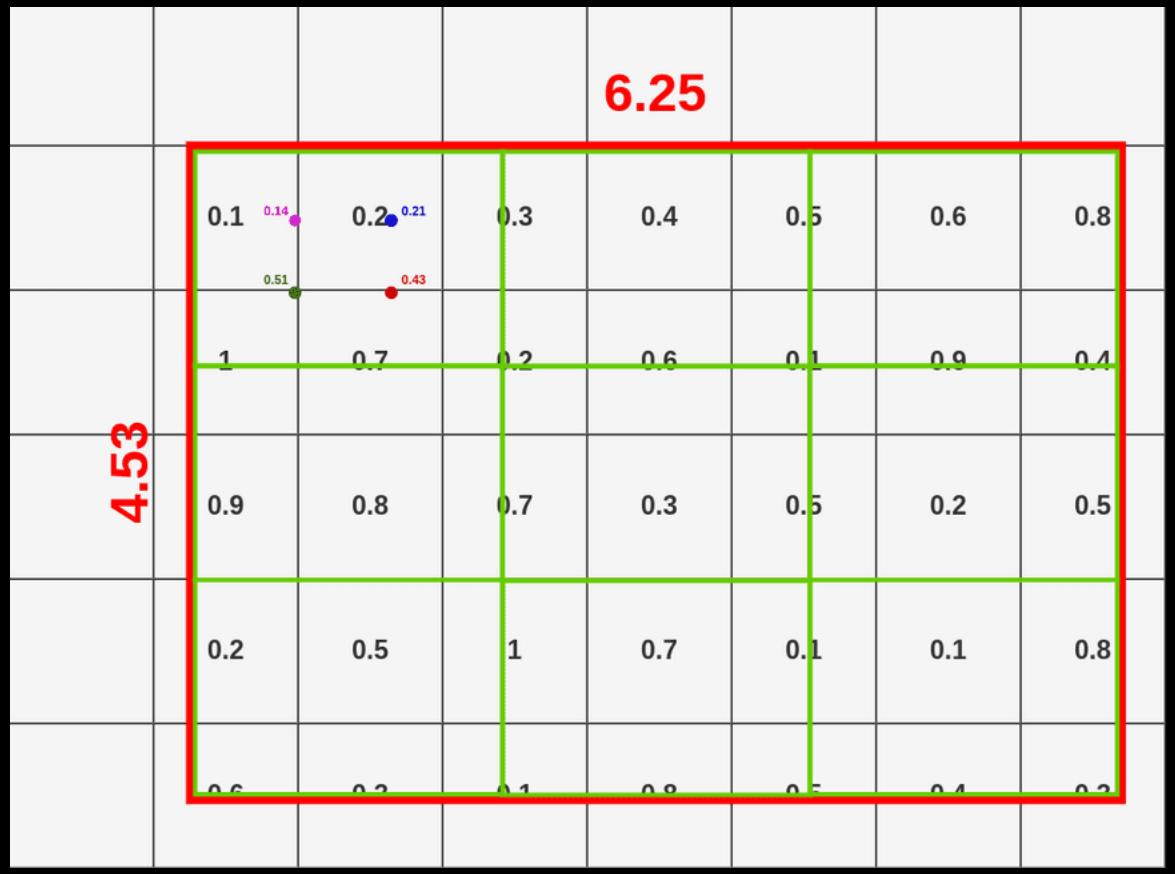
ROIAlign: Pooling *Without* Quantization

Perform pooling on sampled values in each box



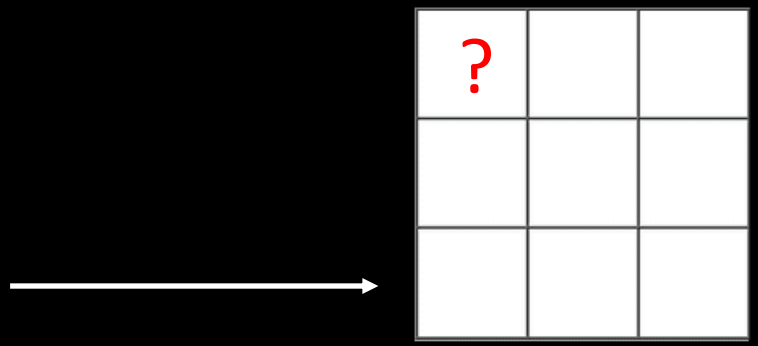
https://erdem.pl/2020/02/understanding-region-of-interest-ro-i-pooling

ROIAlign: Pooling *Without* Quantization



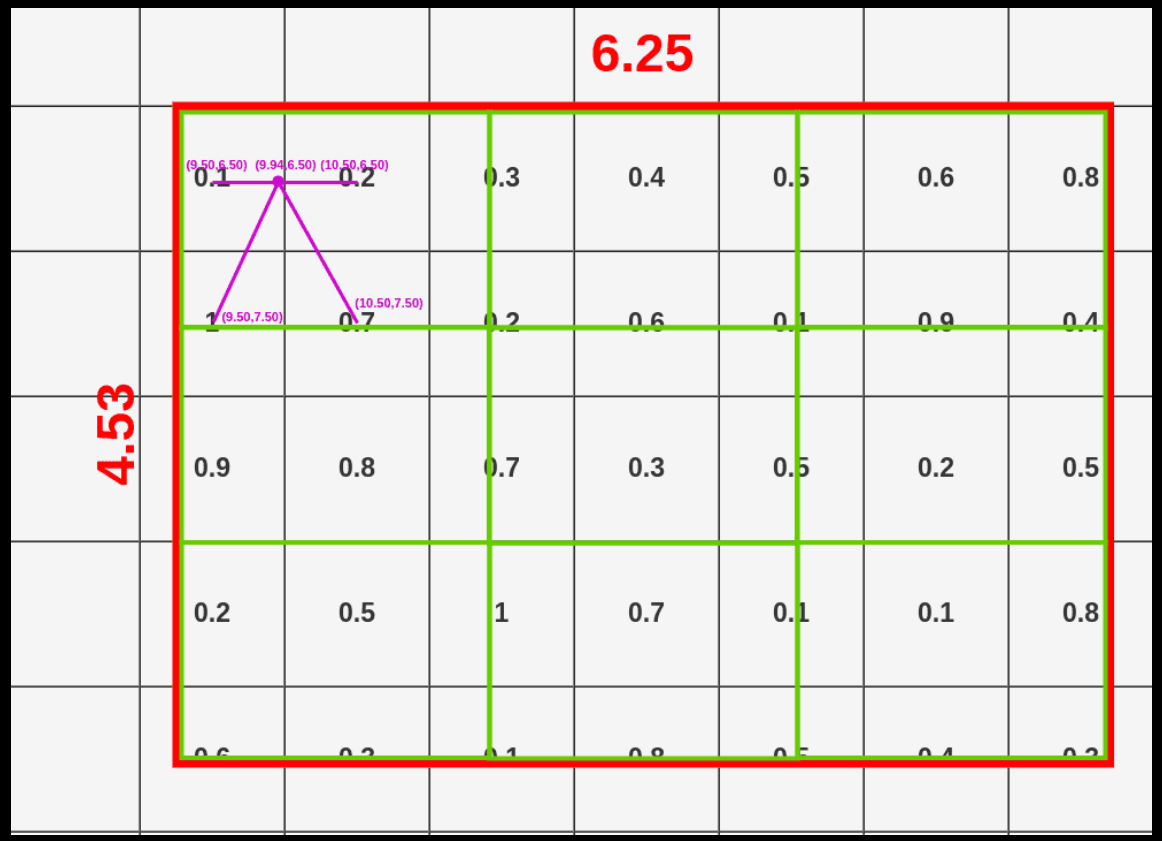
Perform pooling on sampled values in each box
- e.g., max(0.14, 0.21, 0.51, 0.43) = ?

How do we find the four sample values?



https://erdem.pl/2020/02/understanding-region-of-interest-ro-i-pooling

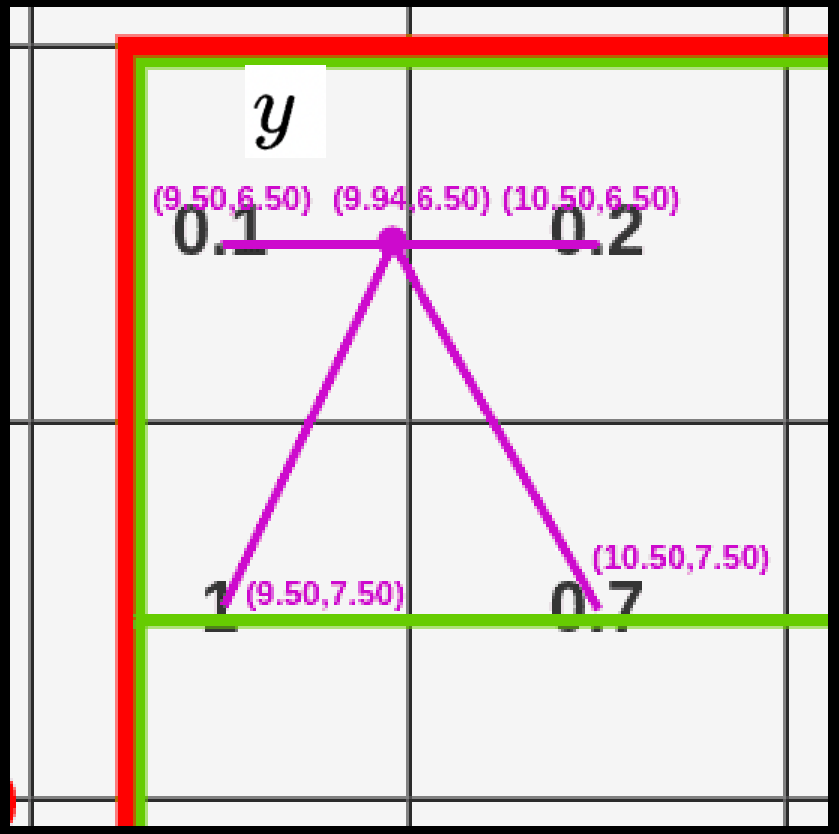
ROIAlign: Pooling *Without* Quantization



Compute each sample value with interpolation between 4 points

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ROIAlign: Pooling *Without* Quantization



Compute each sample value with interpolation between 4 points:

1. Identify sample location

y {
X = X_box + (width/3) * 1 = 9.25 + (2.08/3) = 9.94
Y = Y_box + (height/3) * 1 = 6 + (1.51/3) = 6.50

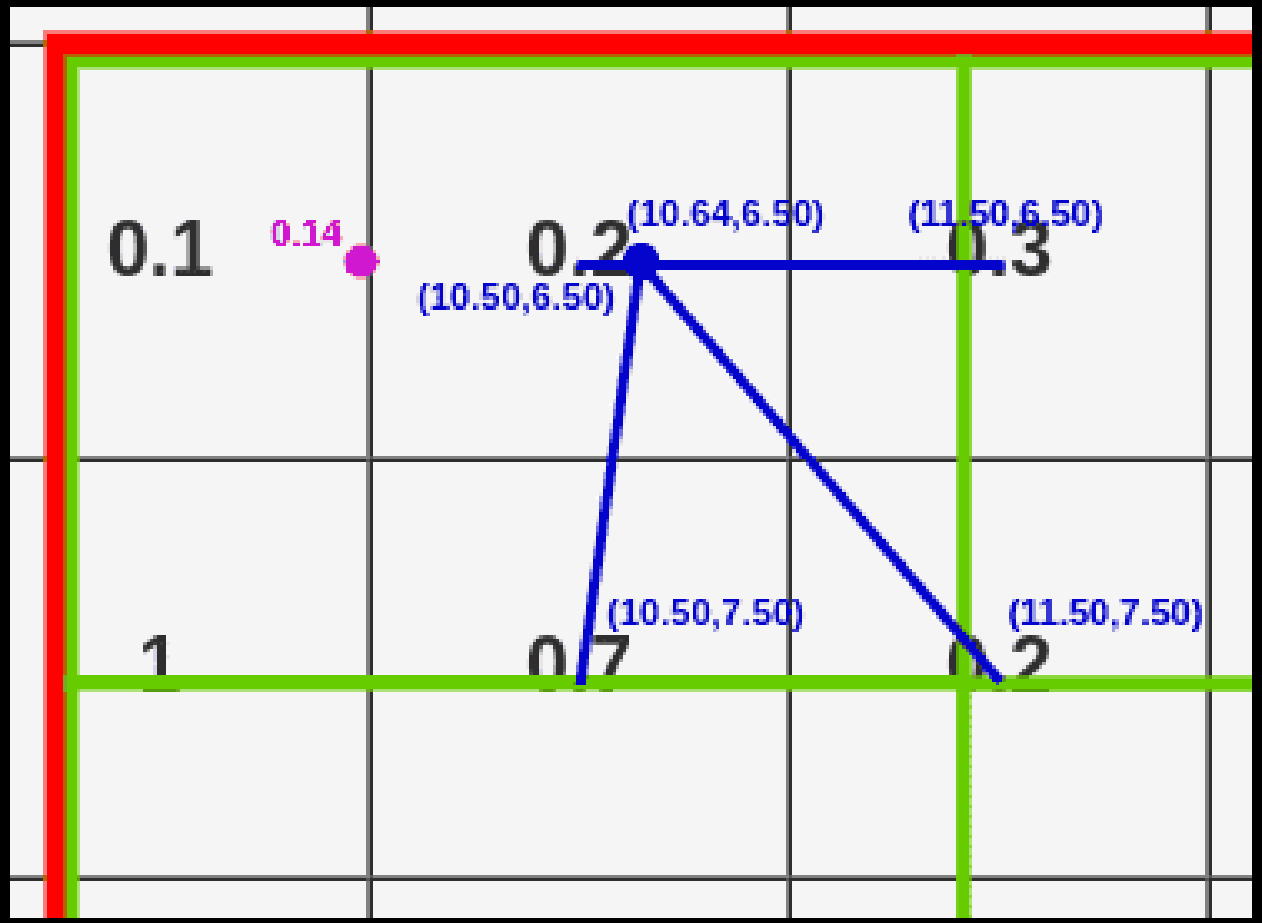
2. Identify 4 points for interpolation, using the middle of each closest neighboring box in each direction
3. Calculate value using bilinear interpolation (= 0.14)

$$P \approx \frac{y_2 - y}{y_2 - y_1}\left(\frac{x_2 - x}{x_2 - x_1}Q_{11} + \frac{x - x_1}{x_2 - x_1}Q_{21}\right) + \frac{y - y_1}{y_2 - y_1}\left(\frac{x_2 - x}{x_2 - x_1}Q_{12} + \frac{x - x_1}{x_2 - x_1}Q_{22}\right)$$

$$\approx \frac{7.5 - 6.5}{7.5 - 6.5}\left(\frac{10.5 - 9.94}{10.5 - 9.5}0.1 + \frac{9.94 - 9.5}{10.5 - 9.5}0.2\right) + \frac{6.5 - 6.5}{7.5 - 6.5}\left(\frac{10.5 - 9.94}{10.5 - 9.5}1 + \frac{9.94 - 9.5}{10.5 - 9.5}0.7\right)$$

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ROIAlign: Pooling *Without* Quantization

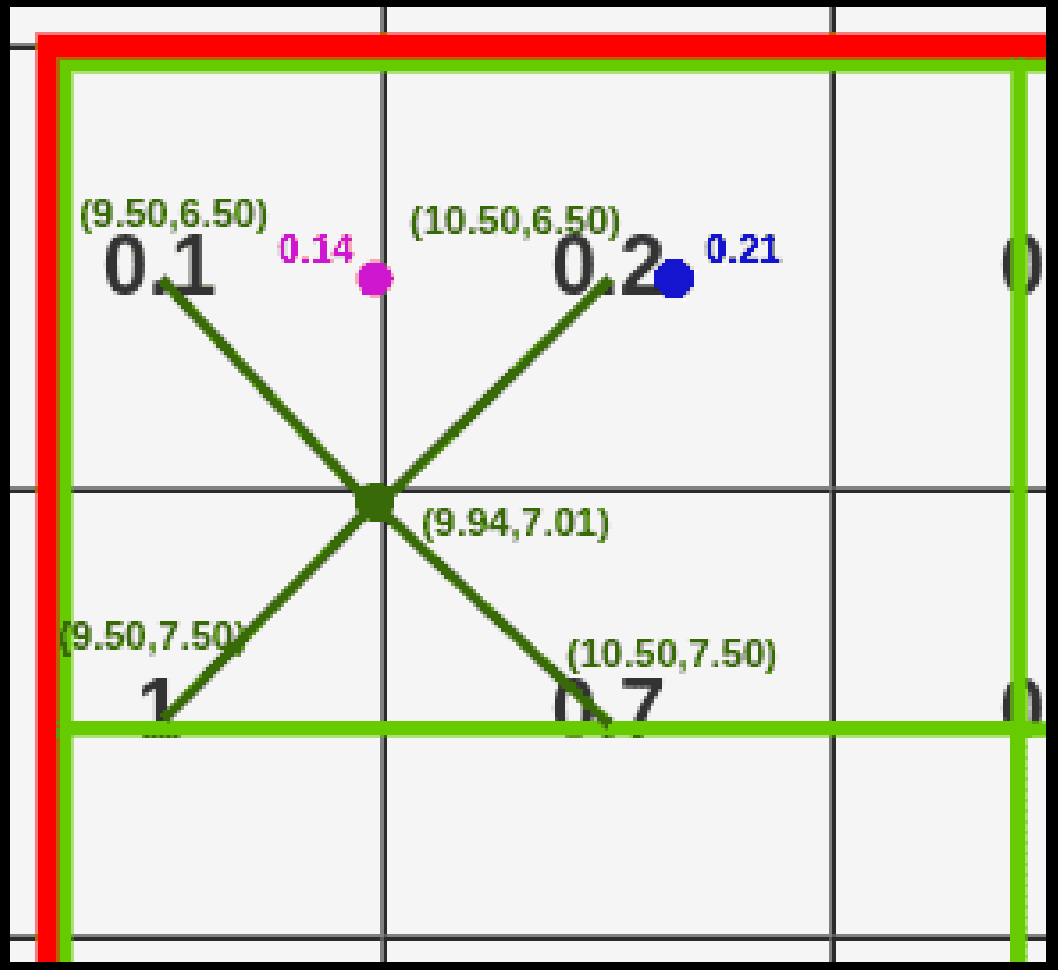


Compute each sample value with interpolation between 4 points:

1. Identify sample location
2. Identify 4 points for interpolation, using the middle of each closest neighboring box in each direction
3. Calculate value using bilinear interpolation (=0.21)

https://erdem.pl/2020/02/understanding-region-of-interest-ro-i-pooling

ROIAlign: Pooling *Without* Quantization

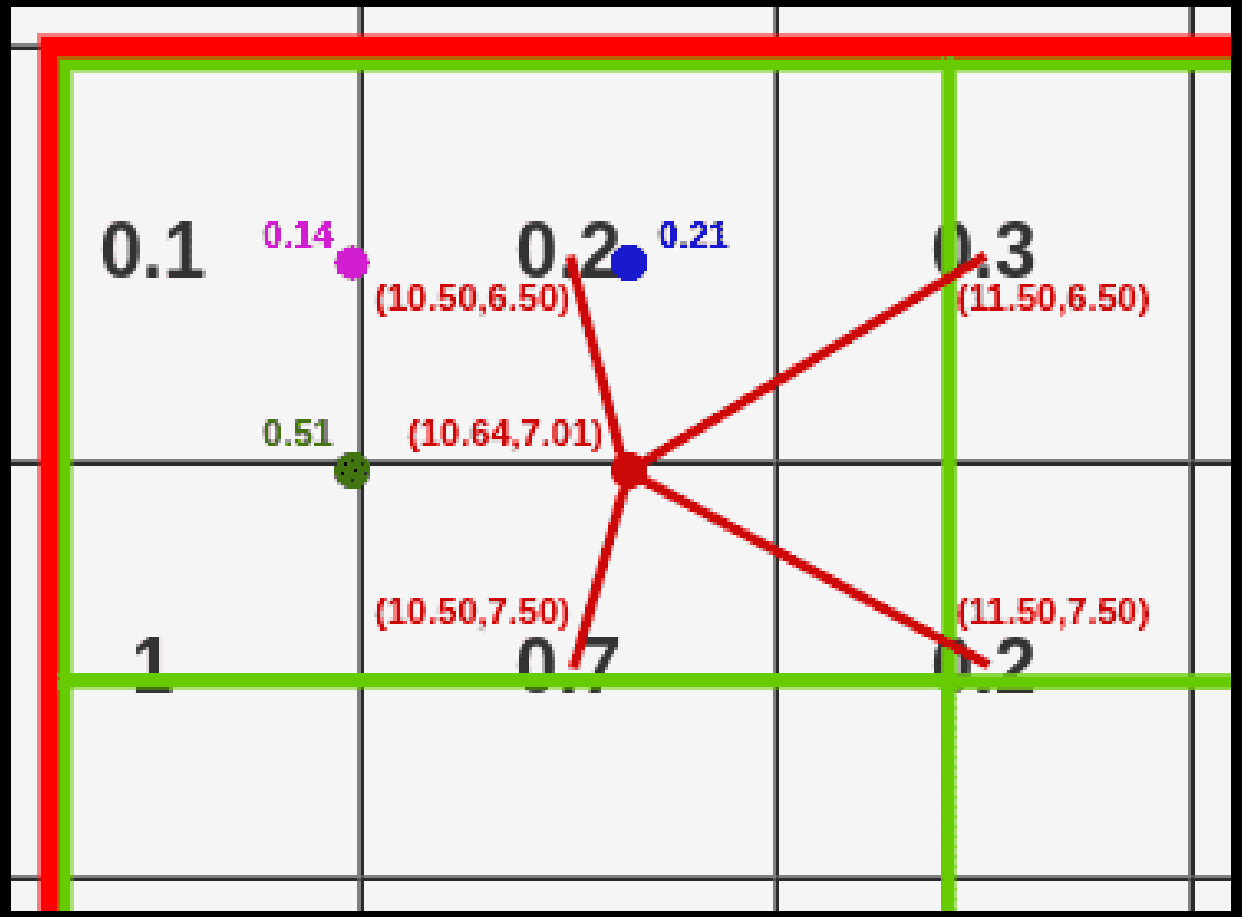


Compute each sample value with interpolation between 4 points:

1. Identify sample location
2. Identify 4 points for interpolation, using the middle of each closest neighboring box in each direction
3. Calculate value using bilinear interpolation (=0.51)

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ROIAlign: Pooling *Without* Quantization

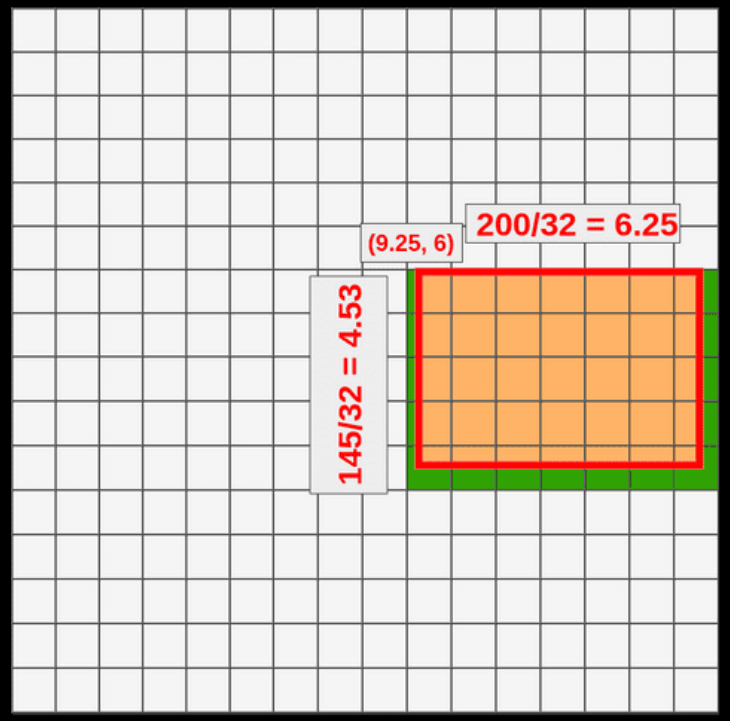


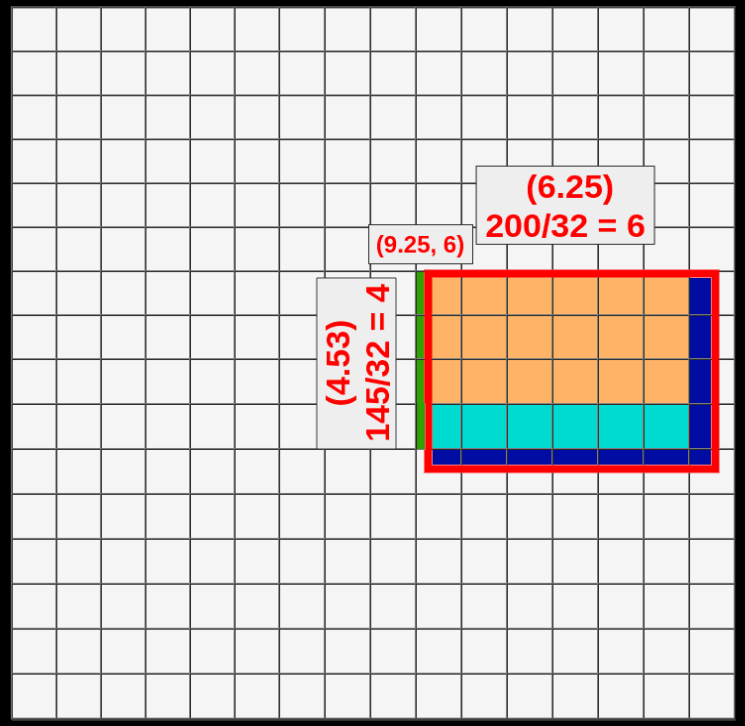
Compute each sample value with interpolation between 4 points:

1. Identify sample location
2. Identify 4 points for interpolation, using the middle of each closest neighboring box in each direction
3. Calculate value using bilinear interpolation (=0.43)

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ROIAlign vs ROI Pooling





Original region on feature map

Both methods add extra image context

Only ROI pooling loses information about the object from the original image

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Today's Topics

- Motivation: when vision asks not just "what", but also "where"
- Redesigning neural networks to predict densely, rather than just classify
- Freeing dense prediction networks from fixed input sizes
- Pivotal insights: pretrained features + design tricks to boost spatial precision
- Programming tutorial

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The End